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Dyson School of Design Engineering

Supporting Semi-Structured Interviewing and Evidence Review

A Formative Design Study of a Controlled Interview-to-Review Workflow

**Master's Project Report
Referral Revision**

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Abstract

Semi-structured interview work includes more than asking questions. Researchers must keep a purposeful line of inquiry, respect a participant's choice not to disclose, and check whether draft statements are supported by the original response. This project asks how a deliberately limited AI-assisted workflow can support that path while leaving protocol design and final interpretation with the researcher.

The study combined literature, three written consultations with research practitioners, a browser prototype, frozen technical and output-quality tests, and a small researcher-review task. The prototype connects protocol state, participant choices, transcript passages, draft statements and researcher decisions. Django stores the records and review process, LangGraph restricts the available interview actions, and GPT-4.1 mini assists with provisional coverage assessment and permitted follow-up wording.

Across 32 registered model runs, the Prototype completed and passed 16/16, compared with 10/16 for the prompt-only condition. Proposition-level checking found 3/46 unsupported proposition instances, all in one edited Prototype item. In a separate within-participant task, three practitioners reviewed one Manual spreadsheet case and one Prototype case. Prototype produced slightly more Gold-aligned decisions and shorter administrator-timed review periods in these six case runs, but it retained one false approval.

The contribution is a bounded interview-to-review design: participant control, visible gaps, linked source passages and the researcher's final decision remain connected. The evidence supports technical conformance and promising review-stage value in this small study. It does not show that the system conducts better interviews, produces better insights, is ready for production, or will be preferred by a wider population.

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1. Introduction

1.1 Motivation

Human-centred design and co-design literature positions user and stakeholder involvement as central to understanding design problems and developing appropriate responses. Sanders and Stappers describe co-design as a shift from treating users as passive subjects towards involving them as contributors to design activity [1], while Kujala identifies early user participation as an important principle for developing usable systems [2]. Qualitative stakeholder research is one way in which design teams understand experiences, needs, constraints and reactions to emerging concepts.

Semi-structured interviews are useful because they combine a planned research focus with the ability to clarify, probe and adapt. DeJonckheere and Vaughn describe the method as a dialogue guided by a flexible protocol and supported by follow-up questions and comments [3]. Kallio et al. argue that careful development of an interview guide contributes to the trustworthiness and plausibility of qualitative findings [4]. This flexibility also creates practical work. The interviewer must keep track of what has already been covered, judge whether another question is justified, avoid repetition, respond to discomfort or refusal, and preserve enough context for later analysis.

Large language models create an opportunity to support parts of this work. They can interpret varied wording, generate a relevant clarification and produce a first-pass structured draft. However, a fluent conversation is not the same as a good research interview, and a polished summary is not automatically usable evidence. Language models can produce unsupported or over-confident text [7]. Any AI-assisted workflow therefore needs a clear division of responsibility: software may support a bounded task, but the participant must retain control over disclosure and the researcher must retain responsibility for protocol design, evidence checking and interpretation.

1.2 Problem framing

The first submission treated scaling as the problem and an AgentDecision trace as the answer. That was too broad and too feature-led. A transcript can already be inspected, and existing interview and analysis products support automated collection, synthesis and source checking. The revised project therefore starts with the work researchers described, not with the novelty of a screen or data object.

The practical problem sits around the interview. Researchers have to design a usable guide, notice what has and has not been covered, preserve refusal and uncertainty, and check draft text against what the participant actually said. Assistance is useful only if it does not hide gaps or transfer more setup and checking work than it saves.

The three written consultations made those needs concrete. C01 found responses difficult to analyse when they were 'unorganised and indirect' and wanted a 'clear logic chain' from source to judgement. C02 described tracking coverage across varied answers and returning to transcripts, audio or notes. C03 had used AI for categorisation but checked its summary against the original interview notes before accepting or revising it. C03 also used facial expression and product-use reaction to guide probing, which places non-verbal studies outside the present text-only case.

The need is therefore not an 'inspectable artefact' in the abstract. It is a less fragmented way to move from response to checked draft while keeping the transcript, unanswered points and final human decision in view. Protocol state, source links and review records are means to that end, not needs by themselves.

1.3 Research question, aim and contribution

The revised research question is:

How might a bounded AI-assisted workflow support design researchers in conducting protocol-guided text interviews and reviewing source-grounded draft evidence, while preserving participant control and researcher judgement?

The aim is to identify when this form of support might be useful, implement one controlled vertical slice, and evaluate its technical behaviour, source faithfulness and review-task value.

The main contribution is an implemented and formatively evaluated handoff from protocol-guided collection to researcher sign-off. Its value proposition comes from observed research work—maintaining coverage, returning to source, preserving uncertainty and approving consequential text—rather than from the existence of an AgentDecision record or a novel interface element.

Three evidence layers support the contribution: three practitioner consultations that framed the problem and use boundaries; a working prototype that separates records, interview control, language assistance and approval; and evaluation that treats technical conformance, semantic faithfulness and researcher-task value as different claims.

The third consultation informs the problem and comparison, not product validation. The separate review task adds direct interaction evidence, but only from three practitioner participants working with synthetic material. Claims about interview quality, whole-workflow workload, representative acceptance and other domains remain open.

1.4 Bounded design case

PurrStone and sensory overload provide a concrete protocol through which the workflow is instantiated. In this report, sensory overload refers to an everyday moment in which a person feels overwhelmed in a noisy, bright, crowded or otherwise sensory-intense environment. The term is descriptive rather than diagnostic. Recent experience-sampling research similarly describes overstimulation as excessive or atypical stimulation that may arise from factors such as crowd noise or unpleasant auditory and visual input [11].

PurrStone is a sensory-support concept used to give the interview a realistic design context. The study does not test whether PurrStone is an effective product or intervention. The protocol asks about one specific overwhelm situation, contextual triggers, bodily or emotional signs, coping and support, reaction to the concept, and public-use acceptability. This bounded episode gives the prototype a defined set of topics and required information without being presented as evidence of cross-domain generality.

1.5 Report structure and claim boundary

Chapter 2 combines the original literature foundation with the referral-stage written consultations and a more cautious account of adjacent systems. Chapter 3 derives user needs, requirements and

operational success criteria. Chapters 4 and 5 describe the final system and the design engineering process. Chapters 6 and 7 separate contextual, technical, output-quality and exploratory researcher-task evidence. Chapter 8 discusses what the combined evidence means for the research question. Chapters 9 and 10 address project management, ethics, reflection, limitations and future work.

Throughout the report, a transcript means the chronological source record of the session. A draft evidence item is a provisional topic-level representation linked to participant messages. An Evidence Record is the collection that remains after explicit researcher review. An AgentDecision is an operational record of a routing action; it is not a disclosure of the model's hidden reasoning. These terms are kept separate so that the report does not use several abstract labels for the same object.

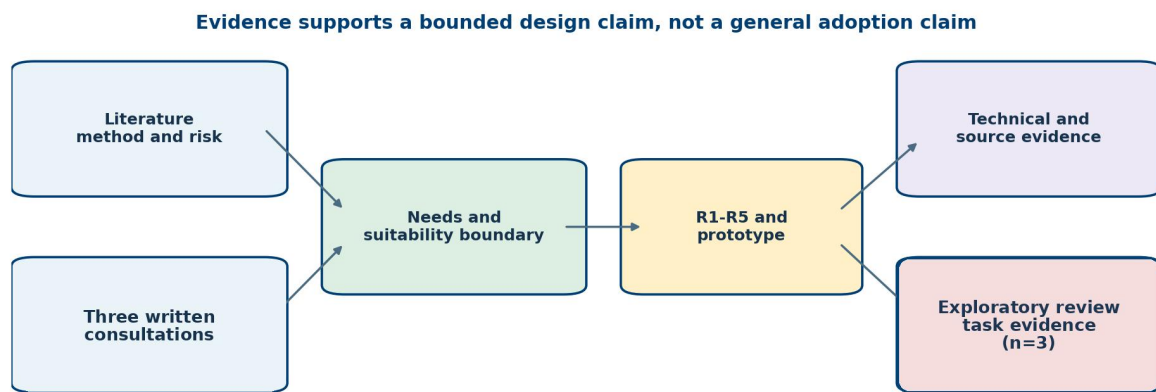


Figure 1. Evidence-to-design logic used in the revised project.

2. Contextual Investigation and Related Work

2.1 Semi-structured interviewing

Semi-structured interviewing is neither a fixed questionnaire nor an unrestricted conversation. It uses planned topics or questions while allowing the interviewer to adapt wording, order and follow-up questions in response to the participant. DeJonckheere and Vaughn describe it as dialogue guided by a flexible protocol [3]. DiCicco-Bloom and Crabtree similarly describe predetermined open-ended questions with further questions emerging from the dialogue [12]. The method therefore involves control decisions as well as language.

For this project, the important implication is that an AI-supported interview cannot be evaluated only by whether its wording sounds natural. It must keep a research purpose in view, avoid unnecessary repetition, respond appropriately when information is incomplete, and respect the participant's decision to skip or stop. The protocol should guide rather than force the interaction. A system that fills every field but pressures the participant or removes uncertainty would not reproduce controlled flexibility; it would convert an interview into a form.

2.2 Interview-guide design and trustworthiness

The interview guide is a methodological control structure rather than a list of prompts. It connects the research question to topics, conversational flow and the evidence needed for later interpretation. Kallio et al. argue that rigorous guide development supports trustworthiness [4]. Castillo-Montoya's Interview Protocol Refinement framework likewise emphasises alignment with research questions, inquiry-based conversation, feedback and piloting [13]. These sources justify making topic purpose and required information explicit before an AI-assisted interview begins.

They do not prescribe the particular interface or data model used in this project. The one-follow-up limit, topic status labels and Evidence Record are project-specific design choices. Their value must be evaluated rather than presented as direct requirements from the literature. Answer sufficiency is therefore defined in Chapter 3 as protocol-element coverage for routing. It is not treated as truth, depth, rapport, analytic saturation or overall interview quality.

2.3 AI-assisted qualitative data collection

AI conversational interviewing has moved beyond simple scripted chat. Recent work reports adaptive questioning, structured guide control and assistance with qualitative data organisation [6, 14, 15, 40, 41]. These capabilities make the project technically plausible, but they also narrow its novelty: guide-following, dynamic probing and automated summaries already exist.

More recent studies underline the need to evaluate the whole interaction rather than the model alone. Wuttke et al. report mode-dependent completion and preference in a large AI conversational-interview study, while SparkMe frames adaptive interviewing as a trade-off among protocol coverage, emergent themes and interview length [44, 45]. Long transcripts also create retrieval risks for language models [8], and established human-AI guidance stresses that system status and available human actions should remain clear [9]. The remaining question here is therefore not whether an LLM can ask follow-ups, but whether collection limits can carry into a usable checking and sign-off task.

The main risk is not only hallucination. Summary and categorisation can also remove context or turn a tentative account into a firmer claim. Qualitative researchers in Yan et al.'s study valued AI assistance but raised concerns about accuracy, consistency, context and trust [43]. Related work calls for transparent validation and researcher feedback loops [16, 43], while the wider hallucination literature shows why fluent wording cannot stand as evidence [7]. Generated material in this project is therefore provisional and must remain linked to participant text and visible gaps.

2.4 Bounded lived experience

Micro-narratives use an AI-guided sequence of focused questions to collect fragments of one lived experience. The system assembles those fragments into a short narrative that the participant can review and correct, giving researchers a participant-confirmed set of situated accounts [5]. This is more than simply asking for a short story: participant confirmation is part of the method.

This project borrows only the bounded-episode principle. It asks about one specific situation, but it does not construct a participant-confirmed micro-narrative. Instead, the application assesses researcher-defined topic coverage, allows at most one follow-up, retains the participant's typed

messages as source material, and later presents provisional evidence to a researcher for Include, Edit or Exclude. The difference is therefore the downstream source-checking and sign-off workflow, not a claim that the project improves the Micro-narratives interview method. Table 2 makes this boundary explicit.

2.5 Contextual investigation method

Stakeholder analysis separated three interests: research practitioners who design and review studies, participants whose consent and disclosure choices must be protected, and ethics or safeguarding oversight. These three interests were selected because they respectively control the research purpose and interpretation, provide and control disclosure of the primary data, and define the permissible conditions under which that data may be collected and used. Direct consultation in this phase covered only the first group. Participant interests informed R2 through interview-method literature, value-sensitive design and the approved ethics boundary [23, 24]; they were not treated as if participants had co-designed the system. The supervisor was not counted as a consultation respondent. The approved written consultation then asked practitioners about current work before introducing a broad hypothetical AI-supported workflow. This ordering was used to reduce solution-led prompting and avoid framing current practice through the existing prototype.

Three de-identified responses were included in the formal analysis. C01 was a Design Engineering graduate with experience in interview-guide design, analysis and review. C02 was a product manager who described designing guides, reviewing original interview material and using findings in product decisions. C03 was an industry research specialist in an automotive context who reported guide design, interview conduct or supervision, analysis of original material, review of interview-derived outputs and use of findings for decisions. All three supplied a participant profile, completion information, consent to de-identified analysis and permission for short de-identified quotation.

The sample was recruited through convenience access and was not intended to be representative. It provides late-stage problem-framing and design-rationale evidence, not saturation, company endorsement, expert validation, co-design, market validation or proof of target-user usability. C03's organisation is reported only at the broad sector level; the company and personal identity are not used in the thesis.

Responses were analysed using a small framework based on the work the project needed to understand: end-to-end workflow, difficult judgements, answer sufficiency, output checking, acceptable software responsibility and adoption conditions. The structure was informed by the framework method's use of a case-by-code matrix [32]. Within each area, statements were coded for the current problem, an observable need or tension, and a design implication. The analysis kept disconfirming material, including reasons why the workflow should not be used. The resulting themes were compared with the literature rather than treated as standalone requirements.

Table 1. Included consultation participants

Code	Broad background	Relevant experience reported	Use in this project
C01	Design Engineering graduate; university context	Guide design, analysis, review and use of findings	Current-practice and conditional-acceptance evidence
C02	Product manager	Guide design, analysis, review and decision use	Current-practice, workload and suitability evidence

Code	Broad background	Relevant experience reported	Use in this project
C03	Industry research specialist; automotive context	Guide design, interview conduct/supervision, analysis, output review and decision use	Industry workflow, text-limit, source-checking and comparison evidence

2.6 Findings from current practice

The first issue was keeping a coherent line without writing an overlong guide. C02 described the difficulty of tracking what had already been covered when participants answered in different ways or anticipated later questions. C01 called responses 'unorganised and indirect'. C03 located the burden even earlier: because users had different expectations, the upfront questionnaire 'tended to become too long'. The design response was a researcher-defined topic purpose and a short list of required information, not automatic question expansion.

The second theme was that sufficient information is contextual rather than a function of answer length. C02 looked for the topics and concrete case context needed by the research question. C01 referred to a complete story and understood rationale while recognising that unwillingness to share more must end probing. C03 used facial expressions during product use to decide whether an answer was clear and whether to probe or move on. This supports protocol-defined coverage and participant control, but also marks a limitation: a text-only system cannot observe non-verbal evidence and should not be presented as a substitute where such cues are methodologically important.

All three contributors returned to source material. C02 located important or ambiguous claims in the transcript and, where available, checked audio or field notes. C01 wanted a 'clear logic chain' from evidence to judgement. C03 had used AI for preliminary categorisation but 'reviewed the AI-generated summary against the original interview notes' before accepting or revising it. This is direct evidence for source-linked draft items and item-by-item review. It does not imply that a citation guarantees support.

The fourth theme was uncertainty and premature closure. C02 argued that a structured record can create the impression that an interview is complete merely because every field has a status. "Not known", "not asked", "participant chose not to answer" and contradictory evidence should remain visible. C03's account similarly required the researcher, rather than the software, to judge responses and derive final conclusions. Status fields therefore describe collection state; they are not scores of the participant or instructions to complete every gap.

The fifth theme was conditional responsibility and refusal. C01 required a clear goal, supervision and human revision. C02 considered assistance plausible only for low-risk, clearly scoped text studies. C03 allowed transcription, preliminary organisation and categorisation, but retained guide design, real-time judgement, probing and final conclusions with the researcher. C03 would refuse the workflow for sensitive personal driving data or without explicit informed consent, required access to a human researcher, and required every draft item to be reviewed. These conditions strengthen the low-risk, text-suitable boundary and participant-protection requirements. Here, General Data Protection Regulation (GDPR) compliance means meeting the legal duties that govern personal-data processing, including lawful and transparent use, data minimisation, security, storage limits and accountability [47]. The prototype was not audited against those duties and is not presented as GDPR-compliant.

Adoption depended on the whole workflow, not the speed of generation. C02 said that guide setup, checking, correction and approval all had to be counted. C03 named the current 'combination of Excel and Word' as the fairest comparator and asked for generation time and researcher revision time to be recorded separately. C03 also raised technical-term accuracy, data security and source-based explanation. The suggested 90% classification threshold is reported as that participant's acceptance condition; it was not added retrospectively to the project criteria because this study did not test it.

2.7 Adjacent systems and the design opportunity

The capability benchmark in Table 2 prevents an inflated novelty claim. Commercial interview tools already offer configurable guides, dynamic probing, branching, transcripts and automated synthesis. Some link outputs to quotations or timestamps. Post-interview tools already support transcript-linked highlights, editable AI suggestions, coding and audit-oriented review [18-21, 34-39]. Protocol control, source citation and human review are established capabilities.

Academic systems lead to the same conclusion. LM-Interview covers guide construction, interviewing, summarisation and scoring [40]. AInterviewer exposes guide control, coverage and refusal classifications, probing decisions and reproducible configurations [41]. Wuttke et al. and SparkMe extend the evidence on scale, mode choice and adaptive utility [44, 45]. This project does not claim to invent AI interviewing, graph control or source grounding.

Table 2 compares these established capabilities with the narrower question retained for the present project.

Table 2. Related approaches and their relevance

Approach	Capability already established	Question retained for this project
Human-led semi-structured interviews	Flexible judgement, rapport, contextual and non-verbal probing.	Remain the reference for sensitive, exploratory, relational or non-verbal work.
Commercial AI-interview platforms	Outset, Listen Labs, Strella and Maze document configurable guides, adaptive follow-up, branching or depth controls, transcripts, synthesis and quotation-linked outputs [18, 19, 34, 36, 37].	AI interviewing, protocol configuration and transcript-linked output are established product capabilities; this project does not claim them as new.
Dovetail, Condens, Looppanel and NVivo	Citation-backed or transcript-linked analysis, editable AI suggestions, coding, source management and audit-oriented review [20, 21, 35, 38, 39].	Source grounding and human review are established; the project examines their connection to collection limits, proposition fidelity and attributable disposition.
LM-Interview	Automated guide construction, interviewing, summarisation, scoring and descriptive analysis [40].	End-to-end automation is established; its published description does not establish the proposition-level anchor and sign-off structure evaluated here.
AInterviewer	Open, customisable guide control, multi-agent classification, refusal/probing decisions, pilot testing, deployment and reproducible configuration [41].	Protocol-guided multi-agent interviewing is established; the present project focuses on the downstream evidence-review handoff.
Recent adaptive-interview studies	Mode-dependent completion and preference at survey scale; optimisation of protocol coverage, emergent themes and interview length [44, 45].	Coverage, participant experience and burden require direct evaluation; model capability alone is not a utility result.
Micro-narratives	AI-guided questions elicit fragments of one lived experience; the system assembles a short narrative for participant review and correction, producing participant-confirmed situated accounts [5].	This project borrows the bounded episode only. It keeps the typed response as source, applies a researcher-defined coverage rule and moves a provisional item into later researcher review; the participant does not confirm an assembled micro-narrative.

Approach	Capability already established	Question retained for this project
Prompt-only condition	Low-setup conversational policy using the same model, protocol, scripted inputs and output/review schema.	Controlled architectural comparator only; not a commercial or academic SOTA product benchmark.

The initial capability benchmark was compiled on 1 August 2026 from public product and research documentation, before the ethics-approved consultations. The cited pages were checked again during final reporting. The table records documented functions, not measured product performance, internal implementation or the absence of undisclosed capabilities. No commercial platform was experimentally tested, so the prompt-only condition is reported separately as an architectural baseline rather than a state-of-the-art product comparison.

The narrower opportunity is the handoff from collection to accountable review. Published descriptions of LM-Interview and AInterviewer do not establish the same proposition-level source-and-sign-off arrangement evaluated here [40, 41]. Vendor pages show capabilities, not independently verified internal rules. The project therefore tests one specific configuration in which topic requirements, a one-probe limit, Skip and Stop, visible gaps, source-linked propositions and Include/Edit/Exclude decisions remain connected.

3. Project Framing and Requirements

3.1 From evidence to requirements

The requirements come from literature, the three practitioner consultations, failures in earlier prototypes and responsibilities to participants. Research-practice evidence mainly informs R1, R3, R4 and R5. R2 is also grounded in participant autonomy and the ethics boundary: consent, Skip and Stop are safeguards even though intended interview participants were not directly consulted in this revision [24]. The recurring needs were to keep a purposeful line of inquiry, let participants limit disclosure, make draft text easier to check, preserve uncertainty and keep final approval with a named researcher.

A technical prototype and the initial related-system benchmark existed before the consultations, so the project was not co-designed. After ethics approval on 6 August, the practitioner accounts changed the referral revision: decision traces became secondary, while participant control, missing information, source checking and attributable review became central. The resulting R1-R5 requirements are summarised below; Appendix C decomposes them into 28 inspectable implementation subchecks.

Table 3. Need-to-design trace

Evidence from literature or consultation	User need or tension	Design response	Evaluation boundary
Flexible protocol and guide alignment [3, 4, 13]; C02's coherent-line problem; C03's guide-design burden	Maintain purposeful coverage without rigid or expanding scripts	Locked topic structure, accumulated topic coverage and one bounded probe	Scripted behaviour only; not guide quality or natural interviewer quality
C01-C03 on willingness, consent and sensitive contexts; C03 on non-verbal cues	Preserve disclosure control and recognise text-medium limits	Consent, Skip, Stop, non-clinical boundary and bounded use case	Tested phrases do not cover every refusal, distress signal or non-verbal cue

Evidence from literature or consultation	User need or tension	Design response	Evaluation boundary
AI uncertainty [7, 16]; C01-C03 source-checking practice	Check each draft against original material	Same-session participant-source links and provisional draft status	A link does not validate interpretation, factual truth or proposition support
C02 on premature closure; C03 on missing/conflicting checks	Keep absence and contradiction visible	Complete, partial, skipped, stopped and not-reached states	Status is protocol coverage, not a quality score
C01-C03 on human control and revision	Retain final responsibility with the researcher	Include/Edit/Exclude with attributable review events	Exploratory n=3 task evidence only; not representative usability or workload proof
C02 on setup-to-approval effort; C03 on the 'Excel and Word' comparator and visible revision time	Avoid hiding work that moves from generation to researcher checking	Compare complete review paths and report generation and revision effort separately	Current task covers review only; whole-workflow time was not instrumented

3.2 Operational definition of Protocol coverage

Protocol coverage is defined narrowly as coverage of the required information for the current protocol topic. It is a routing criterion, not a measure of truth, rapport, insight, credibility, research quality or saturation. A long response can remain incomplete if it omits a required element; a short response can be complete if it covers what the protocol needs. The assessment uses the participant's accumulated messages for the current topic and records both covered and missing elements.

The action rule is bounded. If required information is covered, the workflow records the assessment and moves to the next topic. If information is missing and the probe budget has not been used, the workflow may ask one focused follow-up. If information remains incomplete afterwards, it records the missing element and moves on. Skip and Stop override this rule. A clinical or therapeutic request receives a boundary response and is not converted into ordinary participant evidence.

This definition makes the route testable, but it remains dependent on the researcher's protocol design and the model's semantic assessment. In the historical v9 batch, repeated S2 calls showed that an identical model request could produce different missing-information annotations. The issue did not recur in the current v10.2 MVP batch, but Protocol-coverage judgements should remain inspectable and correctable rather than being presented as objective.

3.3 Design requirements

R1 - Researcher-defined protocol and bounded action control. Before collection, a researcher must be able to create or select a protocol that states the study purpose, ordered topics, primary questions, required information and probe limit. The version used by a session is then locked and identifiable. During the interview, the workflow retains the current topic, accumulated topic evidence and permitted action set, and may only ask one topic-focused follow-up, move on, flag missing information and move on, Skip, Stop or give a boundary response. This operationalises protocol-guided interviewing; LangGraph is the chosen mechanism, not the requirement itself.

R2 - Participant autonomy and boundary handling. The workflow must disclose the study purpose, AI involvement and non-clinical scope before the interview. A participant must be able to Skip the current topic or Stop the session, and those choices must change stored workflow state.

The system must not probe the skipped topic or continue after Stop. Requests for diagnosis or treatment must receive a clear boundary response.

R3 - Source-grounded draft evidence. Every retained or edited draft item must carry at least one valid link to a participant-authored message from the same session and relevant topic. Agent text, system instructions and material from another session must not be represented as participant evidence. Draft items remain provisional until review.

R4 - Visible collection limits. The draft and Evidence Record must distinguish answered, partial, insufficient after follow-up, skipped, stopped and not-reached topics. Missing-information fields should describe protocol elements that remain absent. A topic without participant evidence must not be displayed as approved evidence.

R5 - Attributable human review. The researcher must be able to Include, Edit or Exclude each draft item. Edit and Exclude actions must record reviewer, time and reason, and the final record must preserve relevant sources and coverage limitations. The workflow must keep draft and approved states distinct.

3.4 Requirements and success criteria

Table 4. Requirements, operational criteria and evidence

ID	Operational success criterion	Main evidence	A pass does not prove
R1	Correct protocol/topic state; no more than one topic-focused probe; stored action matches action taken	Frozen protocol, automated tests, S1-S3 and S6	Good interviewing, correct protocol design or LangGraph superiority
R2	Consent gates ordinary interaction; Skip ends topic; Stop ends progression; boundary request receives no advice	Consent route, S4-S6, persistent state	Detection of every natural refusal or distress signal
R3	Retained draft items resolve to same-session participant messages and preserve source meaning	Source-link checks plus supplementary MP and UCR	Factual truth, analytic significance, universal fidelity or a Baseline comparison
R4	Partial, skipped, stopped and not-reached states remain visible; absent material is not completed by inference	S2-S5, review render and missing-information checks	Overall adequacy of the interview or study
R5	Include/Edit/Exclude events retain attribution and change the rendered record as intended	S7-S8 review events plus the exploratory Manual/Prototype task study	Representative usability, objective workload reduction or a valid final interpretation

3.5 Scope and non-claims

The implemented scope is one Sensory Overload / PurrStone text-interview vertical slice. It includes protocol and session records, consent, a participant route, transcript storage, structured draft items, source links, missing-information states, researcher review, an HTML Evidence Record render and operational decision records. It does not include recruitment, scheduling, production identity management, aggregate qualitative analysis, clinical support, a complete safeguarding service or cross-domain validation.

The report does not claim measured scalability, efficiency, better interviews, better insights, equivalence with human researchers, representative target-user acceptance, commercial-platform superiority, universal meaning preservation, zero hallucination, production readiness or PurrStone product effectiveness. These are not rhetorical disclaimers added after evaluation; they define the question the current evidence can answer.

4. System and Agent Design

4.1 Plain-language overview

The prototype is a browser-based workflow used by two human roles. A participant receives a study link, reads the consent and boundary information, and exchanges text with the interview agent. A researcher creates or selects the stakeholder record and protocol, views the saved session, checks draft items against participant messages, and decides what may enter the final record. The system is therefore not only a chatbot: it connects the participant-facing interview to a researcher-facing checking task.

Five linked record types make the workflow checkable. The transcript stores what the participant and agent said. Protocol and session state store the active topic and collection status. Draft evidence items provide source-linked topic summaries for review. Review events store the researcher's Include, Edit or Exclude action. AgentDecision records store the route selected by the workflow, such as follow-up, move next, Skip or Stop; they do not expose hidden model reasoning and are secondary to the transcript and final human judgement.



LLM: coverage and wording · LangGraph: bounded control · Researcher: final approval

Figure 2. End-to-end bounded interview-to-evidence workflow and responsibility boundary. The LLM assesses provisional coverage and words permitted follow-ups; the participant controls disclosure and the researcher owns sign-off.

4.2 Layer separation

The architecture separates records, interview control and language support. Django stores sessions, consent, messages, draft items and review events [25]. LangGraph keeps topic state and limits the

next action [26]. GPT-4.1 mini assesses topic coverage and words a follow-up only after the workflow permits one. The researcher still decides what may enter the final record.

Table 5. Responsibility of each technical layer

Layer	Main responsibility	Implementation anchor	Explicit boundary
Django workflow	Records, routes, consent, transcript, draft items, review and render	models.py: Protocol, InterviewSession, Message, StructuredDigestItem; views.py: review_digest_item(), build_evidence_record_for_session()	Does not decide the next interview action
LangGraph control	Topic state, probe limit, participant-control priority and selected action	langgraph_agent.py: InterviewAgentState, build_interview_graph(), decide_next_action()	Does not validate the research protocol or final interpretation
OpenAI model	Semantic coverage assessment and constrained follow-up wording	langgraph_agent.py: assess_protocol_coverage(); constrained follow-up generation	Does not control protocol order, approve evidence or give clinical advice

Table 5 ties this account to the submitted implementation. It shows where each responsibility is stored or executed. The saved decision records the inputs and route selected by the workflow; it does not expose the model's hidden reasoning. Graph orchestration was chosen to make state transitions testable, not because it is inherently superior to other state-machine implementations.

4.3 Django interview-to-review workflow

Django was selected because the prototype required more than a chat interface. Its relational model layer could persist links among sessions, messages, protocol versions, evidence items and review events, while authentication and server-side validation supported a separate researcher route [25]. A participant accesses a session through a public UUID route; consent and study boundaries precede ordinary questions. The final researcher view brings together the typed transcript, source-linked extracts, collection limitations and review controls. Django therefore supported persistence and auditability, but did not by itself establish good usability or production security.

The review interface separates source, item review and final sign-off. Its fields were included only where they serve R1-R5: protocol purpose, required information and probe limit constrain collection (R1); Skip and Stop preserve participant control (R2); transcript anchors support checking (R3); coverage and missingness keep gaps visible (R4); and Include/Edit/Exclude, reviewer, time and reason make the final disposition attributable (R5). Fields without a requirement or downstream use should be removed in a future simplification. Appendix J shows these stages from sign-in to the final HTML Evidence Record.

4.4 Protocol representation

The Sensory Overload Interview is stored as an ordered set of topics rather than as one large prompt. Each topic has a label, purpose, primary question, required information and maximum probe count. The five participant-facing topics are experience, triggers and signs, coping and support, PurrStone concept reaction, and public-use acceptability. Opening and handoff stages communicate the boundary and prepare the record.

The protocol is versioned and stored with the session so that later review uses the same requirements that governed collection. This prevents a revised guide from silently changing an archived run. The complete topic purposes and required-information fields are reproduced in Appendix A. The system does not determine whether these fields form a good guide; that remains a research-design responsibility and would normally require piloting with the intended population.

4.5 State and routing

LangGraph was used because this interview required an explicit state and a restricted transition set, rather than an unconstrained conversation. In `interviews/langgraph_agent.py`, `build_interview_graph()` links state loading, participant-control detection, coverage assessment, action selection, message generation and persistence. `decide_next_action()` checks Stop, Skip, safety/boundary and discomfort before topic coverage. This makes the permitted route visible and testable, although an ordinary state machine could implement the same principle [17, 26]. Figure 2 shows the runtime order and the human responsibility boundary.

For a sufficient response, the workflow moves to the next topic. For an incomplete response with an unused probe budget, it requests one focused follow-up. If the response remains incomplete after the permitted probe, the workflow records a partial state and moves on. Skip prevents another question on the same topic. Stop terminates further participant collection and prevents post-stop interview records. A boundary request receives a short non-clinical response and does not become normal topic evidence.

The order of these checks is a design value judgement. Participant control has priority over protocol completeness. One probe was selected as a conservative project-specific limit to reduce burden and make the scenarios tractable. It should not be treated as a universal rule for semi-structured interviewing.

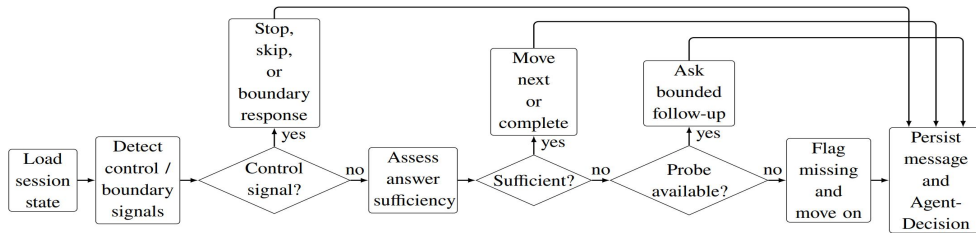


Figure 3. Simplified LangGraph routing logic for one participant turn.

4.6 Model role and failure handling

The frozen evaluation used GPT-4.1 mini through the OpenAI Responses API. It was a pragmatic prototype choice, not the result of a model benchmark. OpenAI documents it as a smaller, faster GPT-4.1 model with low latency, strong instruction following and support for structured outputs [33]. Those properties suited short turn-by-turn coverage assessments and reduced waiting before a participant response or researcher review; its lower cost within the GPT-4.1 family also made repeated development and the 32-attempt formal schedule feasible. The same model ID was used in both conditions to avoid confounding the architecture comparison with a model change.

GPT-4.1 mini is a hosted proprietary model, not an open-source model. Coverage assessment used temperature 0 to reduce unnecessary variation in categorical and structured outputs. Once

application rules had authorised a follow-up, wording generation used temperature 0.2 to allow limited linguistic variation. These values control sampling behaviour, not accuracy, confidence or model authority, and temperature 0 does not guarantee bit-for-bit reproducibility. The model could not approve evidence, change protocol order, diagnose a participant or override Skip/Stop. The formal archive retained 183 real API calls across both conditions.

4.7 Draft evidence and researcher review

A draft evidence item contains a topic, collection status, participant-source link, provisional text and visible missing information. It is a reviewable extract, not a completed qualitative finding. The transcript remains the source, while the Evidence Record documents what the researcher retained, edited or excluded. Figure 4 shows the item-level decision point; the complete interface sequence is provided in Appendix J.

Include retains the draft with its source. Edit requires revised text and a reason while preserving the before/after audit event. Exclude removes the item from the final render while retaining the event. Overall approval is separately gated. In the frozen formal evidence, the include, edit, exclusion and revision-requested paths were exercised. A successful final approval was covered by tests and an earlier live-slice command but not by a frozen formal scenario. Post-approval editing was tested offline, not in the Formal 32. The report therefore separates implementation coverage from formal scenario evidence.

(a) Formal n=3 session and reviewer identity

Output review: NP01-B · Formal 6

Hybrid Meeting Participation Interview · v1 · Transcript-led researcher review

Reviewer: Researcher A

Review pending

← Back to interviews

(b) Claim, source link and review actions

NP01-B-E4 · Breakdown and response

NP01-B-E4 · CANDIDATE EVIDENCE

Covered Pending

SOURCE EXTRACT FROM TYPED RESPONSES

The remote attendee left the meeting after the question was ignored.

CURRENT LINKED TRANSCRIPT SOURCE(S) Participant response 5

Verify or correct source, coverage, and missingness

Include the source extract unchanged when it is relevant and faithful to the linked responses.

Include source extract

Edit before including

Exclude from the Evidence Record

(c) Open coverage and missingness controls

Coverage status for this candidate

Covered

Missing information (one item per line; blank when fully covered)

Reason for correction

Required only when you change a source, coverage status, or missing-information record.

Save source / coverage correction

Figure 4. Researcher review interface from formal session NP01-B-F06, showing the candidate statement, linked source, visible coverage/missingness and Include/Edit/Exclude actions.

4.8 Security and deployment boundary

The prototype uses an SQLite database and local development configuration. Database files may contain transcript, consent and review metadata and must be treated as research data. API keys are stored outside code and excluded from evidence packages; formal scenario runs use synthetic participant inputs. The prototype does not establish production authentication, encryption management, role-based access, retention enforcement, resilience, incident response or GDPR compliance. These are deployment requirements, not completed features. Appendix J documents the implemented browser path but does not provide security or legal-compliance evidence.

5. Design Engineering Process and Implementation

5.1 What counted as an iteration

Two implementation generations are distinguished. Before the referral consultation, the project already had literature review, a related-system benchmark and a technical prototype focused on topic progression, follow-up generation, transcript storage and structured output. After ethics approval on 6 August, three practitioner consultations were completed and analysed. Their evidence then informed the revised question, requirements and evaluation boundary. An iteration is counted only where an artefact or stakeholder input changed the problem framing, responsibility allocation, implementation contract or evidence needed for a claim.

The sequence was therefore: initial benchmark and prototype; ethics approval; practitioner consultation; needs synthesis and requirement revision; workflow and evaluation rework; and a frozen final build. The final version retained the Django foundation and parts of the LangGraph logic, but changed the outcome from an autonomous-looking summary or decision trace to a provisional source-linked item with visible gaps and attributable researcher review.

5.2 Early practice context and dAIsy prototype

An early discussion with Lichen gave preliminary context around AI-supported stakeholder interviewing, follow-up questioning and later analysis [10]. The contact did not continue as a stable organisational brief, co-design relationship or deployment route. The project retained the general problem but moved to an independently buildable PurrStone case. The revised report uses Lichen only as early practice context.

The first working response used Imperial's dAIsy platform [22]. A single prompt set the topic, interview style, limited probing, non-medical boundary and short summary. It demonstrated that an LLM could approximate the conversational surface of a short interview. It did not store an external topic state, distinguish source from draft, support participant and researcher roles, or retain an auditable review event. The lesson was not that prompting was ineffective, but that a prompt alone did not constitute the research workflow being explored.

5.3 Figma workflow modelling

Figma was used to map stakeholder, protocol, session, participant access, output review and evidence status screens before implementation. This clarified that the interview sits inside a wider process and that the researcher and participant require different routes. The models were not user-tested interface evidence. Their role was to make system boundaries visible and support the next build stage.

5.4 Django vertical slice

The Django implementation connected the core records and routes in one vertical slice. The early controller advanced sequentially so that consent, participant messaging, transcript storage and researcher review could be tested together. Once this scaffold worked, the system could expose the central limitation of sequential control: it had no explicit answer assessment, missing-information representation or bounded follow-up decision.

The vertical slice was extended with protocol versioning, separate participant access, structured draft items, source links, review attribution and Evidence Record rendering. These additions made the workflow materially different from the original set of web forms. They also increased setup and review work, a trade-off that was not recognised clearly enough in the original submission.

5.5 LangGraph control and constrained model assistance

The sequential controller was replaced with a graph-managed state transition. The graph loads the topic and protocol snapshot, applies control priorities, calls the model for a constrained coverage assessment where appropriate, and persists both the message and the selected action. Later work limited the model to assessment and follow-up wording so that Stop, Skip, probe limits and routing did not depend on an unconstrained response.

This technical iteration supported testability, but not user value by itself. The frozen scenario suite later checks whether stored actions match the oracle and whether draft items retain sources. It cannot establish whether a design researcher would prefer the interface or whether the system asks good follow-ups in natural fieldwork.

5.6 Referral-stage stakeholder and evidence iteration

Following ethics approval on 6 August, the three written consultations were completed and analysed. C01 and C02 highlighted coherent topic coverage, sufficiency against research purpose, source checking, visible uncertainty and hidden review work. C03 added upfront guide burden, reliance on non-verbal cues, AI categorisation checked against original notes, refusal for sensitive data or absent consent, and Excel/Word as a whole-workflow comparator. This evidence narrowed the use context and supported the revised requirements.

The referral-stage system and evaluation were then reworked around those requirements. Protocol purpose, required information and version identity became explicit; one-probe, Skip, Stop and boundary routes were formalised; draft evidence retained source links and missing-information states; and Include, Edit and Exclude produced attributable review events. Scenario runners, a prompt-only comparator, independent validators, masked review packs and evidence manifests were added to test this chain rather than merely demonstrate it.

This remains formative stakeholder work rather than co-design. The contributors described current practice and responded to a broad hypothetical workflow; they did not operate or approve the final interface. Their evidence supports problem framing, requirements, use boundaries and comparison logic. Utility and usability are addressed separately by the later P01-P03 review task.

5.7 Design choices and alternatives

Django was retained because persistent relational records, authenticated researcher actions and separate public participant routes were already central to the participant-to-review chain [25]. A simple script or front-end-only demo would have reduced build effort but made session identity, provenance and review history harder to represent. The trade-off was additional database, routing and maintenance complexity, and Django's technical controls do not guarantee a usable researcher interface.

LangGraph was retained because the interview required observable state and a restricted transition set: sufficient response, one bounded follow-up, incomplete-after-follow-up, Skip, Stop and boundary handling [26]. A prompt-only model can express the same policy, which is why it was implemented as a comparator; the value of the graph is to externalise state and permitted actions, not to create language ability. The trade-off is greater engineering complexity, while semantic errors such as an inconsistent coverage label can still occur. An ordinary state machine could realise the same principle.

5.8 Final implementation status

The evaluated software was frozen as version v10.2. The formal comparison used one canonical protocol snapshot and 32 registered attempts. Full build and evaluation details are listed in Appendix D. Table 6 states what was implemented, how it was evaluated and what remains incomplete.

Table 6. Final implementation status

System element	Implemented	Formal evidence	Remaining boundary
Protocol setup and versioning	Yes	Frozen protocol used in all 32 attempts	Guide quality not validated
Consent and participant route	Yes	Scripted consent and boundary paths	Natural refusal/distress coverage limited
Coverage and one follow-up	Yes	Prototype 16/16 eligible	Not evidence of better probing
Skip, Stop and boundary	Yes	S4-S6 and persistent records	Indirect or non-verbal cues untested
Transcript, draft and source	Yes	Source-link and MP/UCR checks	A valid link may not support the wording
Include/Edit/Exclude	Yes	S7-S8 and n=3 review task	Human review can still approve an error
Overall approval	Yes	Offline tests and earlier live slice	No successful case in Formal 32
Evidence Record	HTML	26 archived HTML renders	PDF export not validated
Production deployment	No	Not evaluated	Security, retention and legal compliance open

The 28 requirement subchecks are not 28 user tests or a success percentage. They break R1-R5 into 6, 5, 5, 5 and 7 implementation claims respectively. Of these, 24 had direct PASS evidence, two were PARTIAL, one was PARTIAL_MANUAL and one was TESTED_OFFLINE_NOT_FORMAL. Appendix C gives the observable, result and boundary for every subcheck.

The later researcher-validation archive is a separate evidence package rather than a new software release. It contains the procedural pilot, Gold reference decisions used for scoring, six formal condition returns and three final Evidence Records. Human-task findings are tied to those review artefacts; software claims remain tied to v10.2.

6. Evaluation Method

6.1 Evaluation strategy

The original evaluation was mainly a functional test. The revision separates four questions: whether the problem is recognisable in relevant practice; whether the frozen prototype follows its rules; whether reviewed items preserve their source meaning; and whether people can use the review path and see value in it compared with a spreadsheet. This follows design-science guidance

to match each evaluation episode to the property claimed [27]. It also reflects ISO 9241-11's view of usability as an outcome in a specified context, including effectiveness, efficiency and satisfaction [42].

Each layer answers only its own question. Consultation informs need and suitability; Formal 32 and the automated suite test technical behaviour; masked A/B, MP and UCR examine output records; and Validation v1.0.0 tests review-stage effectiveness, experience and desirability with three relevant practitioners. Active review was timed objectively, but six observations cannot establish a causal or whole-workflow efficiency effect.

Table 7. Evaluation layers and status

Layer	Question	Evidence	Status / boundary
Contextual framing	What work, risks and adoption conditions matter?	Three written consultations plus literature	Completed; small, late formative sample
Technical verification	Does the frozen vertical slice execute required controls and records?	197-test suite, Formal 32, 183 API calls and implementation map	Completed within scripted scope
Output faithfulness	Do final reviewed items preserve linked source meaning?	Masked A/B plus two-coder MP/UCR	Bounded supplementary evidence; not universal fidelity
Researcher-review task	Can participants complete and value the review path relative to Manual?	n=3, six matched condition tasks, 60 Gold-scored decisions, ratings and final comparison	Exploratory descriptive evidence; not representative or inferential

6.2 Frozen build and actual-model evidence

The frozen source passed a suite of 197 automated software tests. These tests exercised Django routes and records; consent, Skip and Stop consequences; protocol and session state; graph transitions; draft evidence and source links; Include/Edit/Exclude and review audit; the baseline adapter and scenario runner; metric calculation and HTML rendering; schema, specification and migration checks; and offline paths such as editing after approval. Passing means that the tested code paths followed the project's predefined software rules. It does not show that the interface is easy to use, that interviews are natural, that generated text is always faithful, or that the system is production-ready.

The separate 32-run evaluation provides direct use of the external model: 16 Prototype attempts and 16 prompt-only attempts produced 183 recorded OpenAI Responses API calls. All API responses completed without an API error. Three prompt-only responses were later labelled `parse_error` because the local adapter rejected internally inconsistent content; this was a semantic validation outcome rather than an API outage.

6.3 Scenario design

Eight scripted scenarios represented the main success and boundary paths. S1 supplied complete answers. S2 began with incomplete information and became sufficient after one follow-up. S3 remained incomplete after the follow-up and required a visible partial state. S4 invoked Skip. S5 invoked Stop. S6 tested a non-clinical boundary. S7 exercised researcher Edit and a revision request. S8 exercised Exclude and a revision request. Scenarios S1-S3 and S6 were repeated three times; S4, S5, S7 and S8 ran once, giving 16 attempts per condition. They were chosen to test the main workflow behaviours linked to R1 – R5: complete and incomplete answers in S1 – S3,

participant control and boundary handling in S4 – S6, and researcher Edit and Exclude actions in S7 – S8.

The participant responses and expected action/state checks were fixed before the formal comparison and were not changed after results were seen. The complete 32-attempt schedule was also written in advance. Each registered identity was invoked once and no failed run was retried or replaced. A run entered analysis only if it produced a complete record and passed every applicable predefined check; this status is called formal eligibility. Scenario repetitions and the full oracles are listed in Appendix B.

Table 8. Frozen scenario set

Scenario	Designed condition	Expected workflow consequence
S1	Complete answer	Move to next topic without a follow-up
S2	Initially incomplete, then sufficient	One relevant follow-up, then move next
S3	Still incomplete after follow-up	Record partial/missing information and move next
S4	Skip	End current topic without another probe
S5	Stop	End collection and block later interview actions
S6	Out-of-bound clinical request	Give boundary response without clinical advice
S7	Researcher edits draft	Preserve source, before/after text, reason and reviewer
S8	Researcher excludes draft	Remove from final render and retain audit event

6.4 Prompt-only condition and iteration boundary

The prompt-only baseline used the same GPT-4.1 mini model, protocol, scripted participant inputs and order, topic and probe budget, output fields and downstream review policy as the Prototype. The difference was architectural: it removed the product's LangGraph routing and external action-selection rules and asked one versioned prompt to return coverage, participant control, selected action, covered and missing information, reason and participant-facing wording. It was therefore a controlled comparison of two control strategies, not an ordinary ChatGPT conversation or a commercial product benchmark.

Two versions appear in the report and must not be combined. Historical v9 used Baseline policy v1.0 and produced 1/16 completed Baseline runs. Current v10.2 used policy v3.0 and produced 10/16 completed and eligible Baseline runs in the formal comparison. Code, prompt policy, output schema and local cross-field checks all changed between them, so the difference is design-iteration evidence rather than a nine-run performance gain.

Table 9. Historical and current baseline versions

Version / policy	Result	Role in this report
v9 / Baseline v1.0	1/16	Historical iteration; not pooled
v10.2 / Baseline v3.0	10/16	Current Formal 32 comparison

Historical and current results are therefore reported separately. The current prompt-only condition did not receive the scenario oracle or expected action.

The harness supplied each frozen participant answer only to its registered topic. If the model requested an unregistered follow-up or could not complete that fixed trajectory, the failure was retained. This tests compatibility with the shared scripted workflow; it is not a general evaluation of prompt-only interview quality.

6.5 A/B review

A/B review compared complete Prototype and Baseline outputs from the same scenario. Condition names were removed, A/B position was randomised, and the two coders did not know which system produced A or B. Only pairs in which both runs completed, passed validation and remained formally eligible entered the review. This produced ten pairs. The two coders brought complementary backgrounds: Coder 1 was an MSc Design Engineering graduate with experience in semi-structured interviews and qualitative coding, while Coder 2 was a mechanical design engineer with experience in product development and design review. The preregistered material-generation rule allowed descriptive review with at least eight pairs and identified sixteen as the full target; it was a project-specific execution threshold, not a statistical power calculation.

The coders independently rated protocol coverage (Q1), probing relevance and neutrality when applicable (Q2), transcript grounding and fidelity (Q3), participant clarity and control (Q4), and researcher reviewability (Q5), then chose A, B or Tie [28]. The 1-5 scale used written anchors: 1 meant clear failure, 3 partial or mixed fulfilment and 5 full fulfilment. Six pairs had no genuine probe and were N/A for Q2. The locked independent submissions and disagreements were retained, but complete coder-role and experience metadata were not archived. The ratings therefore cannot be described as verified expert judgement. Appendix H provides the full anchors and data provenance.

6.6 Meaning preservation and UCR boundary

Meaning preservation (MP) asked a simple item-level question: after researcher review, did the final wording still preserve what the linked participant source meant? Two coders independently judged nine final-reviewed Prototype items from S7 and S8 as Preserved or Not preserved. Some items repeated the same source text, so the report gives both nine item instances and five distinct source-text bundles. This is supplementary within-Prototype evidence; no Baseline MP comparison exists.

Unsupported-content rate (UCR) checked the wording more finely. An administrator fixed a 46-row inventory by splitting the same nine items into small factual propositions, following the logic of proposition-level faithfulness checks [30, 31]. The two coders then independently marked each proposition Supported or Unsupported. Initial agreement is reported with Cohen's kappa, a chance-corrected agreement measure [29], before the sole disagreement was resolved. Because the coders did not separately confirm the proposition inventory, 46 is an administrator-frozen denominator rather than coder-consensus segmentation.

6.7 Exploratory researcher-review task

The consultation contributors did not operate the final system, so a separate within-participant researcher-review task was added. Before formal recruitment, a procedural pilot and six-item release check tested the materials, order and data-capture process; pilot data were excluded from the formal $n=3$ analysis. Gold v0.8.0 was then fixed as the scoring authority for 24 topic decisions and 36 evidence-item decisions, giving 60 expected judgements across the six condition tasks. Pilot-01 and Pilot-02 checks were completed by the same HCI MSc graduate and junior UX researcher. A separate MSc graduate with qualitative-research experience checked the reference standards and acted as the Gold reviewer. These details came from the returned pilot and Gold forms, and neither person was included in the formal $n=3$ sample.

Three participants were purposively recruited through personal networks and referrals because their work involved qualitative evidence review. P01 was a service designer with about three years' relevant experience; P02 a Design Engineering PhD student with about two years; and P03 a sustainable-design researcher with about four years. These profiles establish task relevance, but $n=3$ and personal-network recruitment create selection, relationship and social-desirability risks.

Each participant completed one Manual spreadsheet review and one Prototype review across a controlled synthetic protocol (Figure 5). After each condition, 11 post-condition items PC1-PC11 recorded four kinds of information: completion/help/interruption checks; six 1-7 ratings of ease, confidence, source checking, missingness clarity, record accuracy and mental effort; an open comment; and active-review time. The administrator measured time with a stopwatch from the start to the end of the active review, excluding instructions, breaks and questionnaires. After both conditions, ten final comparison items D1-D10 recorded preference, intended real-project mode, willingness, transcript-checking plans, adoption or refusal conditions and requested changes. The 1-7 scales were project-designed to give more detail than yes/no responses; they were not validated psychometric measures, so results are descriptive medians rather than percentages or population scores. Appendix G lists the full instrument structure.

Formal researcher-validation design ($n=3$)

Two conditions per researcher · paired cases within each protocol · near-balanced order

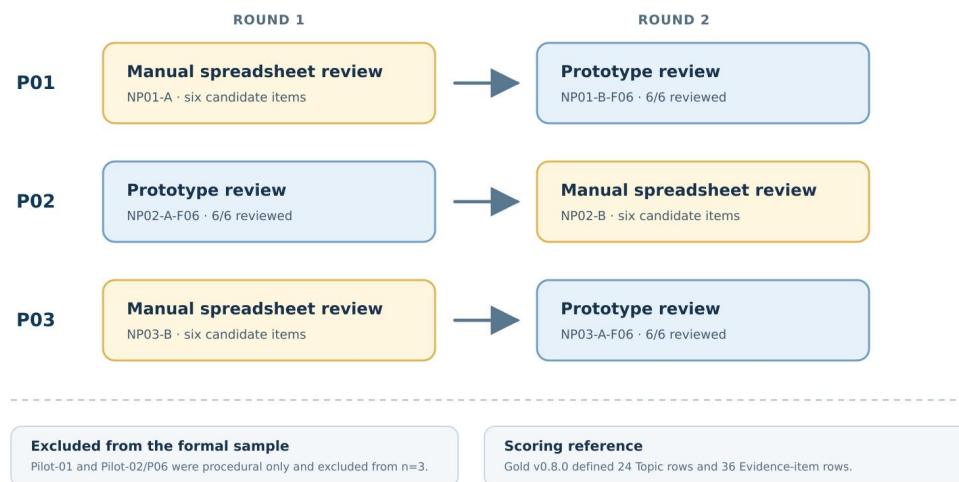


Figure 5. Formal researcher-validation design. Three participants completed paired Manual/Prototype cases in near-balanced order; pilot checks were excluded and Gold v0.8.0 supplied the 60-decision scoring authority.

No inferential test was applied and pilot results were not pooled with $n=3$. Cases were structurally matched but not identical, and the near-balanced order cannot separate interface, participant, case, learning and order effects. The study is therefore direct but exploratory target-user evidence, not a representative usability benchmark.

The study is recorded under Imperial internal ethics record ID 881. The controlled archive retains the approval documentation and the application record. P01-P03 indicated all six consent statements by yellow highlighting and provided names, signatures and dates; the pilot consent and Gold-reviewer role record are separately retained. Identity-bearing records remain restricted, and the thesis reports codes and broad profiles only.

7. Results

7.1 Contextual findings against the requirements

The written consultations did not validate the interface, but they materially changed the interpretation of the design. All three supported protocol guidance, source checking and human revision while rejecting broad automation. C01 required a clear goal, logic chain and supervision. C02 accepted assistance only for low-risk, clearly scoped studies and warned that apparent completeness can conceal unknown, unasked, contradictory or deliberately withheld information. C03 retained guide design, real-time judgement, probing and final conclusions with the researcher, required item-level review, and rejected use for sensitive data or absent informed consent.

C03 also strengthened two practical parts of the project argument. First, reliance on facial expression and observed product-use reaction shows why the research question is explicitly limited to text interviews and why the contribution cannot be generalised to every semi-structured method. Second, Excel plus Word provides a credible current-practice comparator and makes generation time plus researcher revision time part of the workload question. These findings support R1-R5 as plausible requirements and set a whole-workflow adoption criterion: setup and checking work must not outweigh the value of the reviewed record.

The later $n=3$ task addressed one narrow part of that criterion by comparing a manual spreadsheet review with the Prototype review path. It did not convert the three consultations into validation, establish company endorsement or demonstrate general demand for PurrStone.

7.2 Actual-model execution

Formal 32 completed its registered schedule with no replacement attempts. It retained 183 actual OpenAI response records. The 87 MVP and 96 baseline API responses all recorded completed status and null API-error fields. This establishes that the comparison used the real external model path. It does not show service availability over time, natural-session robustness, usability, security hardening or production readiness.

7.3 Current Prototype results and metric definitions

All 16 current Prototype attempts completed and passed the frozen validator. Six automated measures describe why. Action Inspectability (AI) checked that required action-record fields were present (102/102 eligible records). Action Consistency (AC) compared stored actions with the

scenario oracle (86/86 applicable decisions). Source-link Coverage (SLC) checked final retained or edited Prototype evidence items (9/9). Missing-information Visibility (MIV) checked expected gaps (8/8). Participant-control Compliance (PCC) covered one Skip and one Stop scenario (2/2). Topic Coverage Representation (TCR) checked five topics across 16 Prototype runs (80/80). Appendix F reports the full numerators, denominators, units and status boundaries.

Review Audit Completeness (RAC) is the exception. Its frozen calculator returned 12/26 FAIL because the denominator combined 12 executed review actions with 14 pending overall-review placeholders. All 12 actions that actually occurred contained their required audit fields, giving a useful 12/12 diagnostic view, but this does not replace the preregistered 12/26 result. A future version should redefine the denominator before new data are collected.

Table 10. Current v10.2 Prototype result

Measure	Result	Interpretation
Registered attempts	16	Eight scenarios with predefined repetitions
Runner completion	16/16	Every registered current MVP attempt completed
Validator PASS and formal eligibility	16/16	All applicable frozen automatic checks passed
Action Consistency	86/86	Agreement with the scripted action oracle
Missing-information Visibility	8/8	Agreement in the eligible frozen missing-information checks

Historical v9 is retained only as a failure example. Three Prototype runs produced internally contradictory annotations: the participant named braking noise, flashing lights and crowding, but missing_information still listed contextual triggers while the action said sufficient and move_next. These were semantic field contradictions rather than crashes. They did not recur in v10.2, but the model schema still constrains field shape more strongly than cross-field meaning.

The current 16/16 result therefore means that every registered Prototype run in this fixed scripted schedule passed its predefined checks. It does not show that the category-mapping error cannot recur or that the system conducts a better interview.

7.4 Current prompt-only comparison result

The current prompt-only condition completed and passed 10/16 registered attempts. Six failures were retained without retry. All model API responses completed; the failure classes arose from the content of the prompt-only decisions and, in three cases, secondary runner behaviour after the scripted trajectory could no longer finish.

Table 11. Condition completion under the current frozen trajectory

Condition	Registered	Completed and eligible	Retained failures
Bounded MVP workflow	16	16	0
Prompt-only condition	16	10	6

Three baseline runs, S1-R02, S2-R03 and S6-R02, returned covered + move_next while retaining non-empty missing_information. The strict local adapter rejected these internally inconsistent responses. S3-R02, S7-R01 and S8-R01 instead requested an unnecessary final follow-up. Because

these sessions did not complete, the later review steps could not run. These were secondary consequences rather than the original model-decision errors.

The same frozen request could pass in one repetition and fail in another. This supports a narrow finding about nondeterministic rule adherence under the v3.0 prompt, gpt-4.1-mini and this harness. It does not support a universal baseline failure rate or a claim that commercial products are unreliable. Because v9 used a different build and baseline policy, its 1/16 result is retained only as a historical design iteration.

7.5 Independent A/B review result

Ten complete, validated pairs entered masked review. Across the 88 applicable score positions, the coders agreed exactly on 84 (95.5%) and every score was within one point. Overall preferences agreed in 8/10 pairs. Six pairs were unanimously preferred as MVP, two were unanimously tied, and two retained a Baseline-versus-Tie disagreement. No pair was unanimously preferred as Baseline, and no consensus round was performed.

Table 12. Mean quality ratings across coder-pair observations

Criterion	MVP mean	Baseline mean	Interpretation
Q1 Protocol coverage	4.4	4.4	No observed condition advantage
Q2 Probing relevance and neutrality	4.5	5.0	Four applicable pairs only; no MVP advantage
Q3 Transcript grounding and fidelity	4.9	4.9	Equal descriptive ratings among complete pairs
Q4 Participant clarity and control	5.0	4.1	MVP higher in these paired scripted records
Q5 Researcher reviewability	5.0	4.5	MVP higher in these paired scripted records

The result is more informative than the earlier failure-inclusive preference count because both conditions produced complete material for every reviewed pair. It is still descriptive. The small set, unresolved disagreements and coder metadata gaps prevent an inferential superiority claim or a claim of representative target-user preference.

7.6 Meaning preservation and UCR result

The two coders independently judged nine final-reviewed Prototype items. They agreed that eight preserved the participant's meaning and that SF-008 did not; collapsing repeated source text gave four preserved bundles out of five. For the finer-grained UCR check, both coders labelled the same 46 proposition instances. They initially agreed on 45 of 46 labels (97.8%). Cohen's kappa, which adjusts observed agreement for agreement that could occur by chance, was 0.789.

After resolving the single disagreement, 3/46 proposition instances were Unsupported, all in SF-008. The edited item added the interpretive label 'sensory overload' and treated alarms as a cause of feeling trapped and overwhelmed, although the linked source attributed those feelings to noise and crowding. The repeated-text sensitivity result was 3/27, but 3/46 remains the main result. This negative case shows that a valid source link and a completed human edit make checking possible; they do not guarantee faithful wording.

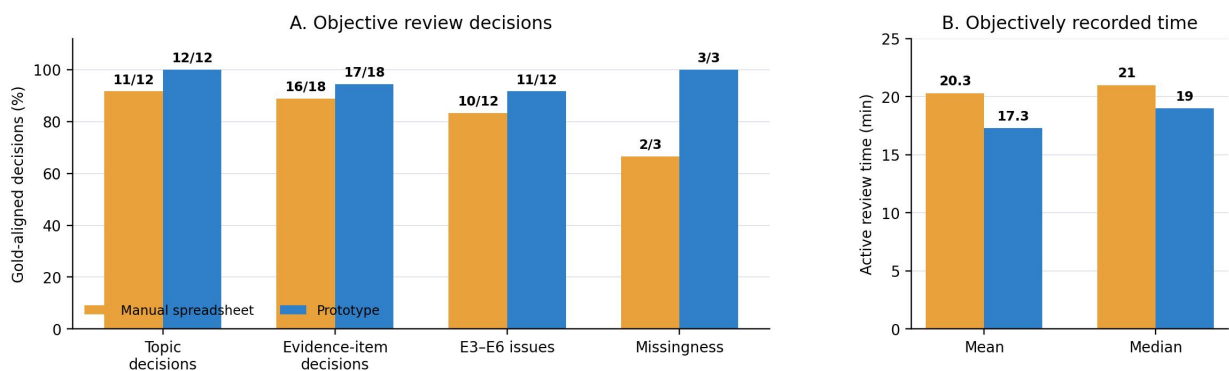
7.7 Researcher-review task result and remaining gaps

All three participants completed both conditions. Against Gold, Manual achieved 11/12 topic decisions and 16/18 evidence-item decisions; Prototype achieved 12/12 and 17/18. Manual's two false approvals were both in P01. For E3, the reviewer accepted a claim that full meeting access had been restored, although the source only showed frozen slides and a switch to phone audio while visual access remained missing. For E6, the reviewer accepted that a clearer turn-taking rule would prevent remote attendees being overlooked, although the transcript proposed the rule without establishing its expected effect; this should have been edited and flagged partial. Prototype's single false approval was P02 NP02-A-E5: the selected source described the same journey but did not support the proposed statement.

Table 13. Formal $n=3$ researcher-review results (descriptive)

Measure	Manual	Prototype	Interpretation boundary
Task completion	3/3	3/3	All assigned condition tasks completed
Topic decisions correct	11/12	12/12	Four per condition and participant
Evidence-item decisions correct	16/18	17/18	Six per condition and participant
E3-E6 issues detected	10/12	11/12	Two issues missed in Manual and one issue missed in Prototype; errors differed
Missingness correct	2/3	3/3	One designated missingness case per run
False approvals	2	1	Manual: two in P01; Prototype: one in P02
PC3 ease, median (1-7)	4	5	Descriptive project-specific rating
PC5 source verification, median	4	6	Higher means easier
PC6 missingness clarity, median	5	6	Higher means clearer
PC8 mental effort, median	6	5	Lower means less effort
Administrator-timed active review, mean	20.3 min	17.3 min	Six observations; stopwatch; review stage only
Condition preference	0/3	3/3	Preference after both controlled tasks

Formal researcher validation ($n=3$; six condition runs)



Descriptive paired results. Cases and order were not fully counterbalanced; one Prototype false approval is retained.

Figure 6. Formal researcher-validation outcomes. Bars show percentages with exact decision counts; timing was recorded by the administrator with a stopwatch. Results are descriptive ($n = 3$), and one Prototype false approval is retained.

Experience ratings were descriptively more favourable for Prototype or equal. Median ease was 4 for Manual and 5 for Prototype; source-verification ease was 4 and 6; missingness clarity was 5 and 6; decision confidence was 5 and 6; and mental effort was 6 and 5. Administrator-timed active-review time averaged 20.3 minutes for Manual and 17.3 for Prototype (medians 21 and 19), with Prototype shorter for all three. Because only six case-specific observations were collected, this is bounded stage-level timing evidence, not a causal or whole-workflow efficiency claim.

All three preferred Prototype for the assigned task, but real-project adoption remained conditional: two chose Hybrid use and one chose Prototype-only. Two would always inspect the full transcript and one would usually do so. All three valued the tool immediately after an interview, during source verification before coding, and at report, audit or handoff stages. The desired role was a pre-sign-off control, not automated analysis or researcher replacement.

Comments explain both value and failure. P01 said that claim, source and intended action were visible on one screen, reducing movement between sheets while preserving transcript checking. P02 described the negative case: the citation concerned the same journey but did not warrant the statement. P03 called the workflow a 'pre-sign-off control' with a path from assertion to anchor to disposition. Participants requested stable ordering, the attributed quotation beside each proposition, and separate labels for 'unsupported assertion' and 'coverage gap'. These post-validation changes were not retrofitted into the evaluated build (Appendix I).

The RAC denominator problem described in Section 7.3 remains part of the frozen technical result. Formal scenarios also lacked a successful overall-approval case and a post-approval reopening case, while source-faithfulness coding covered only a small Prototype universe. The $n=3$ validation adds human evidence but does not change those technical denominators.

7.8 Result summary against the research question

The combined results establish a technically functioning interview-to-review vertical slice and add direct, exploratory evidence about researcher review. Under the frozen scripted protocol, the MVP completed 16/16 runs compared with 10/16 prompt-only runs. Complete-pair review found clearer participant-control and researcher-review records, while protocol coverage and grounding ratings were equal and Baseline probes were rated at least as highly in the four applicable pairs. MP/UCR then exposed one meaning-changing reviewed item containing three unsupported propositions.

In the separate $n=3$ task, both conditions supported completion. Prototype yielded slightly more Gold-aligned decisions (including 17/18 evidence items), one rather than two false approvals, more favourable descriptive ratings, shorter recorded review time for all three cases and unanimous task preference. The single Prototype source-selection error is material: it shows that a traceable interface can still support a wrong approval. Small n , non-identical cases, near-balanced order and personal-network recruitment limit the claim to promising review-stage utility in this task.

The answer to the research question is consequently conditional. A bounded workflow can support protocol-guided interviewing and source-grounded review by keeping protocol state, source anchors, missingness, review actions and decision history in one chain. Its value depends on stable provenance, full-transcript access, visible uncertainty and researcher ownership. Traceability enables scrutiny but neither guarantees fidelity nor replaces qualitative interpretation.

8. Discussion

8.1 Answer to the research question

The answer to the research question is a division of work, not an autonomous interviewer. The researcher defines purpose, protocol and final interpretation. The application keeps topic and review state and retains source history. The model assesses protocol coverage and words a follow-up only when permitted. The participant controls disclosure through ordinary response, Skip and Stop.

The three consultations strengthen the rationale for this division. They converge on the need to connect protocol purpose, contextual judgement, source checking and accountable revision rather than delegate the complete method to AI. C03 makes the medium boundary especially clear: non-verbal observation shaped probing in the described industry project, so the present system supports only a text-suitable subset of practice. The research significance lies in making this responsibility allocation explicit and testable.

The technical evidence shows that this division can be implemented. The n=3 task adds a different finding: all three participants completed both conditions and preferred the Prototype for the controlled review. Their responses centred on seeing assertion, source, missingness and disposition together. This is direct but preliminary evidence of review-stage utility and usability, not evidence that the AI conducted a better interview.

The appropriate conclusion remains bounded. A researcher still has to create the protocol, inspect full context, correct drafts and decide whether an account is analytically usable. The design is most defensible for low-risk, well-scoped text studies in which required information can be stated, uncertainty remains visible and consequential outputs can be checked. Sensitive, deeply exploratory, non-verbal or relational research remains outside the demonstrated use case.

8.2 What the graph-managed workflow contributed

The strongest evidence for the graph-managed implementation concerns external control. It fixed the available actions, stored current topic and probe count, prioritised participant control and linked each transition to persistent records. All 16 current MVP runs completed and passed the frozen validator, compared with 10/16 prompt-only runs. In the complete-pair review, the clearest descriptive differences appeared in participant clarity/control and researcher reviewability rather than protocol coverage or grounding.

This difference should not be read as proof that a graph creates better conversations. The baseline prompt itself was materially improved between v9 and v10.2, and four applicable probe pairs did not favour the MVP. What the current result supports is narrower: externalised state made the predefined protocol path easier to constrain and audit in this implementation. LangGraph is one way to realise that property. An ordinary state machine, carefully written application logic or another orchestration framework could potentially do the same.

8.3 Source grounding and human judgement

Machine checks confirmed that final reviewed items pointed to permitted participant messages, and both conditions received the same high mean grounding/fidelity rating in the ten complete

pairs. The interface also kept Include, Edit and Exclude separate from generation. This implements the practice described by C01-C03: return to the source, distinguish participant account from interpretation, and revise AI categorisation before acceptance.

SF-008 shows why the green checks are not enough. It kept a valid participant link and an attributable human edit, yet both coders judged its meaning changed and proposition coding found three unsupported statements. Research on human-AI decision support likewise warns that simply placing a person in the loop does not ensure critical engagement; safeguards can reduce overreliance but introduce their own usability trade-offs [46]. In qualitative analysis, researchers have also asked for transparency, validation and feedback loops rather than unexamined automation [43]. Traceability creates an opportunity to check; it is not proof that checking was correct.

The Evidence Record is therefore best understood as a reviewed intermediate representation, not the study finding itself. This aligns with C03's allocation of AI to preliminary organisation and categorisation while retaining probing and final conclusions with the researcher. Aggregate coding, pattern development, negative-case analysis and design synthesis remain outside the implemented system.

8.4 Participant control and visible incompleteness

Skip and Stop were not treated as exceptional errors. They changed workflow state and remained visible in the downstream record. This reflects C02's point that absence may mean unknown, not asked, not reached, contradictory or deliberately withheld, and C03's insistence that explicit consent and access to a human researcher remain preconditions for use. A workflow that silently turns these states into blank cells would create false completion.

The scripted controls are still a thin test of autonomy. Natural expressions such as hesitation, partial refusal, distress or a wish to renegotiate consent were not evaluated. A one-probe maximum limits repeated pressure but cannot by itself make an interaction ethical. C03's use of facial expression to decide whether to probe also shows what the text system cannot perceive. In sensitive, exploratory or relational studies, the human interviewer's ability to notice tone, pause, context and discomfort is part of the method. The current workflow should not be used as a substitute in those settings.

8.5 Utility, workload and desirability

Utility, usability and desirability now have direct, although exploratory, evidence. All six tasks were completed; Prototype produced 12/12 correct topic decisions and 17/18 correct evidence-item decisions; ratings were favourable or equal; and all three participants preferred it for the task. In ISO 9241-11 terms [42], this is bounded evidence about effectiveness, efficiency and satisfaction in a specified review context, not a population-level usability result.

The consultation evidence gives the comparison practical meaning. C03 identified Excel for raw-data recording/categorisation plus Word for reporting as the fairest current process and required comparable inputs and outputs. C02 and C03 both argued that setup, generation, source checking, correction and approval must be counted so work transferred to the researcher is not hidden. The

later task compared only the review stage, so it addresses review fragmentation rather than the total cost of the interview workflow.

Adoption was conditional. Two participants chose Hybrid use for real projects and all would inspect the full transcript always or usually. Their conditions—stable anchors, intelligible revision history and clear separation of topical relevance from proposition support—show that usefulness depends on making scrutiny easier without concealing context or researcher responsibility.

Workload remains only partially addressed. Active review was objectively timed and was shorter for Prototype in all three paired participants, while median mental effort was one point lower. However, the task excluded protocol setup, collection monitoring and later analysis, and did not use identical case text. A larger counterbalanced study must measure the full Excel/Word and Prototype workflows before any general efficiency claim.

8.6 Design contribution

The contribution is not a new interview protocol, probing method, transcript, citation feature or review control. It is the tested arrangement of these parts across collection and sign-off. Four lessons are transferable: define routing coverage separately from truth or research quality; keep generated propositions provisional and source-linked; distinguish a topically related source from one that supports the wording; and evaluate technical conformance, semantic faithfulness and human task value with different methods.

The three consultations strengthen the practical origin of this contribution. Their value is not endorsement of the final product but convergence around who must own each judgement: the researcher designs the guide, interprets context, checks sources and owns conclusions; the participant controls disclosure; software may assist with a bounded organisational task. C03 adds the limiting case that non-verbal observation and sensitive-data decisions cannot be absorbed into the present text workflow.

The researcher-review task sharpens the design opportunity. The feature set was not valued because it automated final interpretation; it was valued because it assembled claim, anchor, limitation, disposition and history into an inspectable path. The next design iteration should therefore simplify that path, place verbatim source beside the proposition, keep evidence identifiers stable and separate 'unsupported assertion' from 'coverage gap'.

9. Project Management and Ethics

9.1 Original plan and actual development path

The original report documented weekly notes but did not provide a clear baseline plan, final plan or explanation of changes. Table 14 reconstructs the development path from project artefacts and separates work completed before ethics approval from the later consultation and revision.

Table 14. Planned and actual development path

Phase	Initial intention	Actual output and change
Context and concept	Explore AI-supported stakeholder research and a sensory-overload case	Literature review, PurrStone context and early broad scalability framing

Phase	Initial intention	Actual output and change
Early prototyping	Test whether a conversational system could structure interviews	Lichen/dAlsy experiments and Figma workflow; insufficiently reported as iterations in the first submission
Vertical-slice build	Connect protocol-guided interviewing to researcher review	Django data/workflow layer, LangGraph state control, model-supported coverage and follow-up, review interface
Original evaluation	Demonstrate end-to-end function and compare an agent workflow	Technical tests and thin evaluation; target-user need and utility evidence remained weak
Referral evidence planning, 1-5 August 2026	Review assessor feedback, related work and available comparison options	Initial capability benchmark; draft evaluation plan and ethics materials; no human consultation before approval
Ethics approval and consultation, 6-9 August 2026	Begin approved stakeholder work and refine the problem from practice evidence	Ethics approval on 6 August; three written consultations; revised research question, R1-R5 and use boundaries
System/evaluation rework and reporting, 9-14 August 2026	Test the revised bounded claim and add direct review-stage evidence	Workflow and evaluation updates; frozen 32-run comparison; masked and MP/UCR coding; pilot and n=3 Manual/Prototype task

The largest change was not a new interface feature. It was the removal of claims that the project had not evaluated. The referral work moved scalability and decision inspection out of the research question, treated consultation as problem framing rather than validation, and separated technical verification from target-user evaluation. This reduced the breadth of the claim while retaining the implemented system as a meaningful design object.

9.2 Milestones, evidence control and versioning

The technical evaluation began only after the protocol, scenarios, expected outcomes, prompt-only condition, model, environment and software version had been fixed. The full 32-run schedule was recorded in advance. Each run was attempted once; the six prompt-only failures were kept and no replacement runs were added. The model calls, transcripts, system records, HTML pages, outputs and validation results were retained.

The masked A/B workbooks were returned before the conditions were revealed. MP and UCR kept their original sets of items and all coder disagreements. The implementation map linked each R1-R5 check to its evidence, result, status and what the check could support. This ensured that failed runs and partial results remained visible in the final report.

The researcher-review task was stored separately from the technical evaluation. Pilot-01/02 were retained as procedural checks and excluded from the formal n=3 analysis. Gold v0.8.0 provided the 60 reference decisions used to score the six formal tasks. Validation v1.0.0 preserves the three return packages, Evidence Records, response workbooks, analysis files and deviation log.

9.3 Risk management

Table 15 records the main risks that affected the design and the residual risk after the referral work.

Table 15. Project risk register

Risk	Likelihood / impact	Mitigation used	Residual risk
Broad claims outrun evidence	High / High	Claim matrix, explicit non-claims and separate evaluation layers	Readers may still overgeneralise from a polished prototype
Model output varies between runs	Medium / High	Locked inputs, raw-call archive, validators and retained failures	Semantic annotations may still vary at temperature 0

Risk	Likelihood / impact	Mitigation used	Residual risk
Participant control is missed	Low in scripted cases / High	Explicit Skip/Stop priority, one-probe bound and stored consequences	Natural indirect refusal and distress remain untested
Generated text is unsupported	Medium / High	Same-session source links, two-coder MP/UCR and human review gate	One MP failure and 3/46 unsupported propositions prevent a general reliability claim
External API or data exposure	Medium / High	Research-only deployment, minimal typed inputs and no clinical role	Production security, retention and data residency are unresolved
Review work exceeds benefit	Medium / High	Whole-workflow criterion plus objective active-review timing	n=3 review-stage evidence only; setup, collection and later analysis unmeasured
Late stakeholder access	High / Medium	Used three consultations only for formative framing and disclosed sample limits	Requirements were not co-designed, representative or saturated
Evaluation package is misunderstood	Medium / Medium	Frozen denominator, N/A rules, implementation map and claim boundaries	Baseline completion result remains easy to overstate
Human-data governance is misreported or exposed	Medium / High	Restricted archive; ethics ID 881; signed consent; deviation log	Identity materials and the detailed approval records cannot be published

9.4 Ethical position and participant autonomy

The project ethics application defined an adult, non-clinical study and disclosed AI-supported text interaction. Approval is recorded under Imperial internal record ID 881, with the approval documentation and application record retained in the restricted package. Consultation contributors are reported as C01-C03 and review participants as P01-P03; synthetic task materials do not reproduce real participant disclosures.

P01-P03 consent records indicate all six statements by yellow highlighting and include names, signatures and dates. The procedural pilot participant checked the study process and was excluded from n=3; the Gold reviewer checked the reference standards used for scoring; A/B and MP/UCR coders judged technical outputs; and P01-P03 completed the formal Manual/Prototype task. These roles were not pooled as one participant sample. Signed forms, identities and the detailed approval records are not reproduced in the public thesis.

All public reporting uses participant codes and aggregate or paraphrased responses; signed forms and names remain restricted. C03 is reported only by broad role and sector, without personal or company identification. A real deployment should minimise requested personal data, state retention and deletion periods, restrict researcher access, disclose external model processing and preserve the ability to Skip or Stop without penalty.

The task study and consultation evidence do not justify using the system in clinical, sensitive, non-verbal or deeply exploratory research. C03 explicitly identified sensitive personal driving behaviour and absent consent as refusal conditions, and P01-P03 still expected full-transcript access and human sign-off. Any future study that changes participant group, task, data flow or researcher interaction should be checked against the approved protocol and receive an amendment or new approval where required.

9.5 Ownership and use of generative tools

I designed the project framing, prototype workflow, protocol representation, evaluation specifications and final argument, and I remain responsible for the accuracy of the report. Generative AI supported coding, debugging, language and formatting during the referral revision. I checked its outputs against the source repository, frozen evidence, consultation forms and assessor feedback; AI-generated wording was not treated as empirical evidence, and missing participant or coder information was not inferred.

10. Reflection, Limitations and Future Work

10.1 Personal reflection

I spent too much of the early project treating the AgentDecision trace as the contribution. Because I could build and test it, I started to describe it as if researchers had asked for it. The consultations forced me back to their actual work: keeping a line of inquiry, deciding whether an answer has enough context, checking draft text and deciding whether the extra review is worth doing. I should have begun there and treated the technology as one possible response.

The second change was in how I understood evaluation. The 197 tests can show that a route or record behaves as specified; they cannot show that a researcher wants the workflow. The later task exposed interface friction, conditional adoption and the continuing importance of the full transcript. Its favourable results still do not justify a general utility claim. Verification and validation have to be planned as different pieces of work, not placed under one final heading.

The failures were uncomfortable but useful. A response can be valid JSON and still contradict itself. A draft can have the right source link and still change the meaning. SF-008 did both the work I had designed for it - it kept a source and an edit history - but still contained three unsupported propositions. That result changed 'traceable' from a quality claim into a condition for checking.

The review task made that lesson practical. I had designed around the existence of links; participants described the work of checking. The P02 Prototype error showed that a same-session link can be semantically wrong, while requests for an inline quotation and distinct failure labels showed how the decision path should change. I also learned that consent, recruitment, timing and version records are part of the research product, not administrative detail.

If I repeated the project, I would plan stakeholder work, technical verification, human evaluation and evidence governance together from the start. I would involve intended users before committing to the architecture, freeze denominators and failure rules, counterbalance equivalent cases, instrument every workflow stage and compare the full Excel/Word process. The main lesson is to build from a demonstrated decision burden and keep every claim no larger than its evidence.

10.2 Limitations

The first limitation is stakeholder engagement. The three contextual consultations were late, written and recruited through convenience access. They represented useful but narrow academic, product and automotive-industry perspectives; they were not representative, did not reach

saturation and did not constitute co-design, company validation or expert approval. They improved problem framing and exposed text-medium and adoption limits, but could not undo the fact that most architecture decisions had already been made.

The second limitation is the researcher-review sample. Only three purposively recruited practitioners completed the matched task. Their documented backgrounds establish relevance, but recruitment through personal networks/referrals creates selection, relationship and social-desirability bias. Cases were structurally matched rather than identical, with Manual first for two participants and Prototype first for one; participant, case, learning and order effects cannot be separated.

The third limitation is measurement. PC3-PC8 and D1-D10 were project-specific rather than validated psychometric scales. Active review time was objectively recorded, but only for six condition runs and excluding protocol setup, interview monitoring, breaks, questionnaires and later analysis. Descriptive differences cannot establish statistical significance, causal efficiency or total workload reduction.

The fourth limitation is ecological validity. Formal technical inputs and researcher-review materials were synthetic and used one technical Sensory Overload protocol plus three controlled review protocols. They did not reproduce rapport, non-verbal observation, unexpected disclosure, long answers, participant misunderstanding, messy transcripts or the consequences of a real interview. The review task tested source verification and sign-off, not whether the AI conducted a good interview.

The fifth limitation is comparison validity. The prompt-only condition changed from v1.0 to v3.0 between historical and current builds, so 1/16 and 10/16 cannot be pooled. The A/B review contains ten complete pairs and cannot isolate LangGraph from adapter and control-policy differences. The Manual/Prototype task also compares two whole review configurations rather than one isolated interface feature. Although C03 identified Excel plus Word as the fairest current-practice comparator, the task operationalised the Manual condition primarily through a spreadsheet and did not measure the complete Word-report stage.

The sixth limitation is source faithfulness and implementation coverage. MP covered nine nominal/five unique MVP items and UCR 46 proposition instances/27 unique texts, all within MVP. One reviewed item failed MP and contained all three unsupported propositions. Frozen RAC was 12/26; wrong-session and wrong-topic injections were not separately archived; a successful overall approval and formal post-approval reopening were absent; and the Evidence Record was HTML rather than a validated PDF export.

The seventh limitation concerns evidence storage. Some files from the researcher-review study contain names or signed consent records and therefore cannot be published. The technical evaluation and researcher-review task were archived separately. I later found that three archived records used an inconsistent full software-version reference. The correction was recorded without changing the study data or reported results. Future releases should link each software version directly to its evidence.

10.3 Prioritised future work

The first priority is one preregistered, counterbalanced whole-workflow study with a larger, independently recruited target-user sample. It should compare the complete Excel/Word process with the revised Prototype; include protocol setup, collection handoff, source checking, correction, approval and later use; retain participant-level timing and errors; and measure task accuracy, workload, navigation problems, conditional adoption and transcript checking. Messier transcripts and natural variation should be introduced gradually.

The second priority is to implement the changes raised by the $n=3$ task on a future version. Evidence identifiers should remain stable and ordered; the relevant quotation and surrounding context should appear beside each proposition; and the interface should distinguish an unsupported assertion from a coverage gap. A topical-match warning may prompt review but must not make the final decision. These changes require a new evaluation because they were not part of the tested build.

The third priority is to close known technical gaps: enforce cross-field semantic checks, add formal tests for wrong-session and wrong-topic source links, include successful approval and post-approval reopening in the frozen suite, and repeat MP/UCR over a larger set of items confirmed by the coders. Only after these steps should another domain or system design be compared. Sensitive or deeply exploratory research should remain human-led unless substantially different safeguarding and methodological evidence are developed.

11. Conclusion

This project moved from a broad claim about scaling qualitative insight to a narrower question about protocol-guided text interviews and source checking. Three written consultations identified practical needs around guide length, coherent coverage, visible uncertainty, participant control and human review. C03 added a clear boundary: non-verbal observation can guide probing, AI categorisation still has to be checked against original notes, and sensitive data or absent consent are reasons not to use the system.

The resulting Django, LangGraph and OpenAI prototype connects a participant-facing interview to a researcher-facing review. In the frozen technical evaluation, all 16 MVP runs completed and passed the validator, while 10/16 prompt-only runs completed. Ten complete pairs showed equal descriptive ratings for protocol coverage and grounding, no MVP advantage for probing, and higher MVP ratings for participant control and reviewability. Supplementary MP found one meaning-changing reviewed item and UCR identified 3/46 unsupported proposition instances.

The researcher-review task adds direct human-facing evidence. Three relevant practitioners completed both conditions. Prototype achieved 12/12 correct topic decisions and 17/18 evidence-item decisions, compared with 11/12 and 16/18 for Manual; administrator-timed review time was shorter for all three; and all preferred Prototype for the task. Yet Prototype retained one false approval, two participants chose Hybrid use for real work, and all expected continued transcript

inspection. The result supports a promising pre-sign-off review path, not general effectiveness or causal efficiency.

The practical contribution is a clearer division of responsibility. The model can assist with topic-coverage assessment and follow-up wording; application logic can hold state and source history; the participant controls disclosure; and the researcher defines the protocol, checks context and owns the conclusion. The system's strongest value is not autonomous insight generation. It is a shorter path from a draft assertion to its source, limitation, decision and edit history.

That path still has limits. A source link does not guarantee fidelity, a related quotation may not support the exact wording, and a polished record cannot replace the transcript, non-verbal fieldwork or sound research governance. The final claim is therefore a responsibility-aware design case with evidence-backed next steps, not a claim that AI has solved qualitative interviewing.

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Use of Generative AI Tools

I acknowledge the use of OpenAI ChatGPT/Codex to support coding and debugging, language refinement, structural checking and formatting review during the development and referral revision of this project. Imperial's dAIsy platform was used during early prompt-prototype work. OpenAI's API was used by the final prototype for the bounded semantic-assessment and follow-up tasks described in this report. I reviewed the resulting code, prose and evidence against the frozen source material and remain responsible for the final content.

I confirm that the submitted work is my own. In particular, the project framing, design rationale, prototype scope, implementation decisions, evaluation interpretation, final argument, and responsibility for the submitted work are my own. All AI-assisted outputs were reviewed, edited, verified, and incorporated only where appropriate.

Source Repository

A curated public release snapshot of the implementation and evaluation evidence is available at <https://github.com/Spongebosheep/Jingyang-Liu-02252219-Master-Project.git>. The frozen internal v10.2 evidence package records the assessed source identity as commit 35c930d92119d47560879496902e928daf20ac02. The accompanying evidence package records environment, protocol and content hashes so that the report can be checked against the submitted artefacts. The earlier v9 commit is retained as historical iteration evidence and is not pooled with the current result.

Appendix A. Sensory Overload Interview Protocol Summary

The implemented protocol asks about one bounded everyday episode and the participant's reaction to the PurrStone concept. It is not a diagnostic interview. Required information is used only to assess protocol coverage and route the interaction.

Table A1. Protocol topics and required information

Topic	Primary purpose	Required information for routing	Boundary
Experience	Establish one concrete episode	Situation, place or activity, and what the participant was trying to do	Do not infer a condition or diagnosis
Triggers and signs	Understand what contributed to overwhelm and how it was noticed	Relevant sensory/contextual trigger and bodily, emotional or behavioural sign	One focused follow-up at most
Coping and support	Understand what the participant did and what helped	Action or coping response and perceived effect	Do not give clinical or therapeutic advice
PurrStone concept reaction	Gather response to the design concept	Initial interpretation, expected use and perceived value or concern	The study does not test product effectiveness
Public-use acceptability	Explore social and situational fit	Acceptable context, concern or reason for non-use	Respect refusal and avoid pressure to justify it

Participant-control rules are evaluated before ordinary semantic routing. Skip ends the current topic and records it as unavailable for evidence. Stop ends collection and prevents later interview actions. A clinical or therapeutic request receives a non-clinical boundary response and is not used as participant evidence. Otherwise, a response that meets the Protocol-coverage threshold moves to the next topic. A response below the threshold permits one focused follow-up; material that remains incomplete is recorded as partial before progression.

Appendix B. Frozen Scenario and Oracle Summary

Table B1. Formal scenarios and repeated runs

ID	Repetitions	Core input condition	Main automatic oracle
S1	3	Complete answer	Covered; move next; no follow-up
S2	3	Initially incomplete, then complete	One relevant follow-up; accumulated coverage; move next
S3	3	Still incomplete after one follow-up	Partially covered; visible missing information; move next
S4	1	Explicit Skip	Skip action; no later decision for same topic
S5	1	Explicit Stop	Stopped state; negative POST creates no later research record
S6	3	Clinical or therapeutic request	Non-clinical boundary; boundary text not used as evidence
S7	1	Researcher edits one draft	Before/after text, reason, reviewer, time and source retained
S8	1	Researcher excludes one draft	Item absent from final evidence; attributable event retained

The same scenario/repetition set was run once for the bounded MVP and once for the prompt-only condition, producing 32 registered attempts. The complete schedule was recorded before execution. Every identity was invoked exactly once; formal eligibility required runner completion and all applicable checks in the frozen validator result. Failed runs were retained without retry or replacement.

The prompt-only condition received the same model family, protocol content, scripted participant-answer sequence, run budget, output schema and review policy. It did not receive graph state or AgentDecision routing. If it selected a topic for which the next frozen participant answer had not been written, the harness stopped. This rule protects input comparability but limits the result to fixed-trajectory compatibility.

Appendix C. Implementation Status and Claim Map

The five requirements are decomposed into 28 checkable implementation claims: six for R1, five each for R2-R4 and seven for R5. The identifiers show hierarchy rather than score (for example, R1.2 is the second check under R1). PASS means direct evidence was found; PARTIAL means implementation or archived evidence was incomplete; PARTIAL_MANUAL means the check depends on bounded human coding; and TESTED_OFFLINE_NOT_FORMAL means the path was tested outside the 32 formal runs. These statuses must not be converted into a 24/28 success rate because the subchecks are not equal statistical units.

ID / requirement	Subcheck	Observed result / status	Claim boundary
R1.1 R1 Protocol-bounded progression	Protocol is locked and versioned before formal execution.	All 16 MVP attempts completed under the same locked protocol snapshot. STATUS: PASS	Supports the frozen Sensory Overload protocol only; not cross-protocol reliability.
R1.2 R1 Protocol-bounded progression	Protocol snapshot and cryptographic identity are saved per counted attempt.	All 32 attempts record the same frozen protocol snapshot hash and source commit. STATUS: PASS	Hash identity does not imply deterministic LLM output.
R1.3 R1 Protocol-bounded progression	Every participant step saves one inspectable decision record.	The 16 completed MVP attempts passed control-record cardinality and field-completeness checks at executed steps. STATUS: PASS	Scripted formal scenarios, not unobserved production traffic.
R1.4 R1 Protocol-bounded progression	Recorded high-level actions follow the frozen oracle.	All 16 MVP attempts passed the frozen automatic action/annotation checks; MVP AC=86/86 and MIV=8/8. STATUS: PASS	Frozen scripted-oracle agreement; not natural-conversation quality or deterministic model behaviour.
R1.5 R1 Protocol-bounded progression	Covered and missing-information annotations match the frozen oracle.	All 16 MVP attempts passed the frozen automatic action/annotation checks; MVP AC=86/86 and MIV=8/8. STATUS: PASS	Frozen scripted-oracle agreement; not natural-conversation quality or deterministic model behaviour.
R1.6 R1 Protocol-bounded progression	Follow-up is bounded to at most one probe before progression.	S2 and S3 repetitions preserved the one-probe bound and cumulative progression behavior. STATUS: PASS	Only the frozen topic sequence and scripted inputs were tested.
R2.1 R2 Participant controls	Research content is gated on consent.	Consent-gate checks passed in all 16 MVP attempts. STATUS: PASS	Demonstrates the scripted consent path, not legal or ethics approval.
R2.2 R2 Participant controls	Skip is recorded as an explicit participant-control action.	S4-R01 passed the explicit skip-action oracle. STATUS: PASS	One scripted skip scenario.
R2.3 R2 Participant controls	A skipped topic is not pursued again in the same session.	No later same-topic decision was recorded after the S4 skip. STATUS: PASS	Does not establish behavior for every possible conversational paraphrase of skip.
R2.4 R2 Participant controls	Stop terminates collection and blocks post-stop record creation.	Session status remained stopped; the post-stop negative test created no new research records. STATUS: PASS	One scripted stop and one archived post-stop negative probe.
R2.5 R2 Participant controls	Control consequences remain visible in downstream records.	S4 and S5 limitation states were visible in the review and archived Evidence Record HTML. STATUS: PASS	Actual PDF export is not established by these HTML renders.

ID / requirement	Subcheck	Observed result / status	Claim boundary
R3.1 R3 Source-grounded evidence	Every retained evidence candidate has at least one participant-source link.	Machine source-link validation passed for the nine final-reviewed included items; all 26 completed runs retain archived source-bearing outputs where applicable. STATUS: PASS	Link validity is distinct from semantic fidelity; supplementary MP found one not-preserved item.
R3.2 R3 Source-grounded evidence	Linked sources belong to the same session and participant role.	Source-grounding checks passed for all topic records in the 16 completed MVP outputs. STATUS: PASS	Machine linkage correctness is distinct from semantic faithfulness.
R3.3 R3 Source-grounded evidence	Linked sources are constrained to the same protocol topic.	Same-topic grounding checks passed across the completed MVP corpus. STATUS: PASS	Does not prove that text faithfully preserves the source meaning; MP/UCR addresses that separately.
R3.4 R3 Source-grounded evidence	Invalid sources are rejected by explicit negative tests.	Boundary-message exclusion was directly tested. No separately archived injected wrong-session and wrong-topic rejection attempts were identified. STATUS: PARTIAL	Do not claim a complete negative-test matrix until wrong-session and wrong-topic injections are archived.
R3.5 R3 Source-grounded evidence	Semantic fidelity and unsupported content are human-coded separately from link validity.	Both coders agreed on all nine MP labels: eight preserved and SF-008 not preserved. UCR initial agreement was 45/46 (97.8%), kappa=0.789; final resolved unsupported content was 3/46 (6.5%). STATUS: PARTIAL_MANUAL	Supplementary within-MVP analysis only; nine reviewed items, no Baseline UCR, and no separate coder-confirmation round for the 46-row inventory.
R4.1 R4 Coverage visibility	Every protocol research topic receives exactly one downstream coverage record.	All 16 completed MVP outputs passed exact topic-record cardinality. STATUS: PASS	Stopped baseline attempts did not reach downstream output and are not treated as completed Evidence Records.
R4.2 R4 Coverage visibility	Answered topics show their evidence-candidate and source state.	Normal-path answered topics matched the frozen topic outcomes and source constraints. STATUS: PASS	Machine linkage/coverage is distinct from meaning preservation; SF-008 was not preserved.
R4.3 R4 Coverage visibility	Partial coverage retains explicit missing information.	S3 repetitions preserved partial coverage and visible missing-information limitations. STATUS: PASS	Specific scripted unresolved-answer pattern.
R4.4 R4 Coverage visibility	Skipped, stopped, and not_reached topics remain explicit and are not evidence candidates.	All frozen unavailable-topic outcomes and candidate boundaries passed. STATUS: PASS	One skip and one stop scenario.
R4.5 R4 Coverage visibility	Coverage and limitation state are consistent across saved data, review page, and Evidence Record render.	Visibility checks passed in saved data and 26 archived Evidence Record HTML renders; report screenshots are post-run derived views. STATUS: PASS	HTML rendering and screenshots only; no actual exported PDF is present.
R5.1 R5 Researcher review and audit	Include/approve actions create final reviewed item states.	S7/S8 scripted item actions passed; eight Include actions are stored as approved/is_included=true, one item is edited, and one is excluded. STATUS: PASS	Scripted authenticated reviewer actions; no independent outcome-quality judgement.
R5.2 R5 Researcher review and audit	Edit requires changed text and a reason.	The edited text, reason, and final evidence consequence passed the frozen S7 oracle. STATUS: PASS	One scripted edit example.

ID / requirement	Subcheck	Observed result / status	Claim boundary
R5.3 R5 Researcher review and audit	Edit audit retains reviewer, timestamp, previous text/status, new text/status, and source links.	S7 retained the attributable edit event and preserved source links into final evidence. STATUS: PASS	Audit integrity is shown for the archived test, not external tamper resistance.
R5.4 R5 Researcher review and audit	Exclude removes an item from final evidence while retaining its audit event.	The scripted exclusion path passed and the attributable event remained in the archive. STATUS: PASS	One scripted exclusion example.
R5.5 R5 Researcher review and audit	Overall review saves decision, note, reviewer, time, source-check flag, and non-approval rationale.	S7 and S8 saved attributable revision_requested decisions, notes, timestamps, and source-link checks. STATUS: PASS	Meaning-preservation and protocol-boundary flags remained false because the runner did not substitute for independent human judgement.
R5.6 R5 Researcher review and audit	Approval is gated until all required checks are complete.	The formal scenarios exercised revision_requested, not a successful approval. A separate live-slice command and tests cover the approval gate, but this is not a formal positive-path scenario. STATUS: PARTIAL	Do not claim the positive approval path is formally evaluated.
R5.7 R5 Researcher review and audit	Any post-approval content change reopens approval.	The 197-test offline suite includes a direct post-approval edit test and passed under a synthetic local Git wrapper; no frozen formal scenario exercises the transition. STATUS: TESTED_OFFLINE_NOT_FORMAL	Claim implementation/test coverage only, not formal-scenario performance.

Appendix D. Evaluation Evidence Identity and Claim Boundary

Table D1. Frozen build and evaluation identity

Field	Recorded value
Software commit	35c930d92119d47560879496902e928daf20ac02
Worktree	Clean in archived formal-run check
Environment	Python 3.11.15; Django 5.2.14; LangGraph 1.2.0; OpenAI package 2.37.0
Model	gpt-4.1-mini through the Responses API
Technical attempts	32: 16 MVP and 16 prompt-only
Technical protocol	One locked canonical snapshot across counted attempts
Researcher-task package	PS-TRV-2026-01; Validation v1.0.0; 12 August 2026
Researcher-task runs	n=3 participants x 2 conditions = 6; three controlled synthetic protocols
Task scoring authority	Gold v0.8.0; 24 topic plus 36 item decisions
Version boundary	The technical evaluation and researcher-review study were archived separately.

Table D2. Evidence-to-claim boundary

Evidence	Supports	Does not support
197/197 offline tests	Internal consistency of tested code paths	Target-user value, production reliability or security
Formal 32 and 183 API responses	Actual-model scripted execution	Natural-session robustness or long-term availability
16/16 MVP vs 10/16 prompt-only	Compatibility with the v10.2 frozen trajectory	General LangGraph or commercial-platform superiority
Masked A/B, MP and UCR	Bounded reviewability and source-faithfulness findings	Universal fidelity or representative user preference
Three written consultations	Formative needs, risks and suitability conditions	Usability, demand, saturation or co-design
n=3 Manual/Prototype task	Exploratory task accuracy, experience, preference and objective review-stage timing	Population effects, causal or whole-workflow efficiency, interview quality or representative acceptance
Validation v1.0.0 manifest, hashes and deviation log	Integrity and limitations of archived human-study artefacts	A new software release or public disclosure of restricted governance records

Appendix E. Consultation Analysis Framework

The referral-stage consultation analysis used a small matrix to prevent the system description from becoming the coding framework. The current-practice questions were analysed before the hypothetical AI-support questions.

Table E1. Analysis framework

Area	What was looked for	How it informed the project
End-to-end work	Preparation, conduct, analysis, checking and decision use	Defines the workflow rather than only the chat interface
Difficult judgement	Repeated, high-effort or consequential decisions	Identifies where assistance might be useful or risky
Sufficiency	Observable reason to probe, move on or accept absence	Informs protocol-element coverage and participant-control rules
Output checking	How a draft is compared with source material	Informs source links, limitations and human review
Responsibility	Tasks software may support and tasks it should not own	Defines model, workflow, participant and researcher boundaries
Adoption/refusal	Conditions for use, comparison and non-use	Narrows the intended context and exposes workload as an open criterion

C01-C03 were retained because their forms included de-identified background, relevant research activity, completion information, consent and quotation preference. The forms were not converted into numerical scales. Statements were compared for convergence and tension, and negative conditions were retained in the requirements and Discussion. C03's suggested 90% classification agreement remained a participant-proposed acceptance condition rather than a retrospective project threshold.

Table E2. Consultation contribution matrix

Code	Distinct evidence added	Resulting design or claim implication
C01	Coherent story, clear logic chain, supervision and human revision	Support source-to-judgement visibility and reject autonomous interpretation
C02	Purpose-based sufficiency, multiple forms of absence, low-risk suitability and whole-workflow workload	Support coverage states, visible missingness, bounded use and workload boundary
C03	Guide-design burden, non-verbal probing cues, AI categorisation checked against notes, consent/data refusal and Excel/Word comparison	Support researcher-defined protocol, text-only boundary, item-level checking and fair-comparison logic

Appendix F. Formal Result Tables

Table F1. Counted technical attempts and eligibility

Analysis role	Counted	Completed	Formal eligible
MVP	16	16	16
Prompt-only	16	10	10
Total	32	26	26

Table F2. Current v10.2 automated metric results

Metric	Denominator / unit	Result	Status and boundary
Action Inspectability (AI)	Eligible action records	102/102	PASS; required fields present
Action Consistency (AC)	Applicable action-oracle comparisons	86/86	PASS
Source-link Coverage (SLC)	Final retained/edited MVP evidence items	9/9	PASS; link validity, not semantic support
Missing-information Visibility (MIV)	Expected missing-information checks	8/8	PASS
Participant-control Compliance (PCC)	One Skip and one Stop scenario	2/2	PASS
Topic Coverage Representation (TCR)	16 MVP runs x five topics	80/80	PASS
Review Audit Completeness (RAC)	12 actions + 14 pending placeholders	12/26	FAIL; executed-action diagnostic was 12/12

Table F3. Supplementary meaning-preservation summary

Unit	Reviewed	Preserved	Not preserved	Boundary
Nominal final reviewed items	9	8	1	MVP S7/S8 only
Unique source-text bundles	5	4	1	Duplicate texts collapsed

Table F4. Supplementary UCR summary

Measure	Result	Reporting boundary
Frozen proposition instances	46	Nine final-reviewed MVP items
Unique proposition texts	27	Duplication-sensitivity denominator
Initial exact agreement	45/46 (97.8%)	Before resolution
Cohen's kappa	0.789	Chance-corrected agreement
Final unsupported	3/46 (6.5%)	Resolved within-MVP result
Unique-text sensitivity	3/27 (11.1%)	Repeated text collapsed

Table F5. A/B review summary

Measure	Result	Boundary
Complete eligible pairs	10	Minimum eight met; recommended sixteen missed
Applicable paired score positions	88	176 raw condition scores across two coders
Exact score agreement	84/88 (95.5%)	Four one-point differences
Overall preference agreement	8/10	Six unanimous MVP; two unanimous Tie
Unanimous Baseline preference	0/10	Descriptive, not inferential superiority

Table F6. Researcher-review objective results

Measure	Manual	Prototype
Formal condition tasks	3	3
Topic decisions correct	11/12	12/12
Evidence-item decisions correct	16/18	17/18
E3-E6 issues detected	10/12	11/12
Missingness correct	2/3	3/3
False approvals	2	1

Table F7. Researcher-review ratings and choices

Measure	Manual	Prototype	Boundary
Ease median	4	5	1-7; higher is easier
Source verification median	4	6	1-7; higher is easier
Missingness clarity median	5	6	1-7; higher is clearer
Mental effort median	6	5	1-7; lower is less effort
Administrator-timed active review, mean	20.3 min	17.3 min	Six observations; stopwatch; review stage only
Condition preference	0/3	3/3	After both controlled tasks
Real-project mode	-	Hybrid 2; Prototype 1	Conditional intended mode

Appendix G. Researcher-Review Task Protocol

Study PS-TRV-2026-01 used a within-participant comparison of Manual spreadsheet review and the Prototype review path. Pilot-01/02 were two procedural checks completed by the same HCI MSc graduate and junior UX researcher; both were retained as procedural evidence but excluded from the formal $n=3$ analysis. A separate MSc graduate with experience in semi-structured interviewing and qualitative analysis acted as the Gold reviewer, checking the reference standards against the transcripts, case specifications and scoring protocol; this role was also excluded from the formal sample. P01-P03 had approximately two to four years of relevant research experience and were purposively recruited through personal networks/referrals. Cases were structurally matched rather than identical.

The task evaluated review, not AI interview quality. Participants judged four topic states and six evidence items per condition, corrected sources or wording where needed, handled missingness and produced a final record. Gold v0.8.0 supplied the scoring authority. PC1-PC11 were completed after each condition; D1-D10 were completed after both. The administrator recorded PC11 active-review time with a stopwatch, excluding instructions, breaks and questionnaires. Tables G3-G4 list the instrument items so the codes are not treated as unexplained measures.

Table G1. Condition allocation

Code	Broad profile	Protocol	Round 1	Round 2
P01	Service designer; ~3 years; prior AI transcription/summarisation	NP01	Manual NP01-A	Prototype NP01-B
P02	Design Engineering PhD; ~2 years; limited AI research-tool use	NP02	Prototype NP02-A	Manual NP02-B
P03	Sustainable-design researcher; ~4 years; routinely checks AI drafts	NP03	Manual NP03-B	Prototype NP03-A

Table G2. Researcher-review measures

Construct	Measure	Reporting rule
Task effectiveness	Gold-aligned topic and item decisions; issue detection; missingness; false approvals	Counts only; no inferential test
Usability	Completion, help and interruption; PC3-PC8 ratings	Project-specific 1-7 scales; medians
Workload	PC8 mental effort and administrator-timed active-review time	Participant-level descriptive timing; not causal or whole-workflow efficiency
Desirability	Preference, real-project mode, willingness, transcript checking and feature retention	Conditional choices and paraphrased reasons
Design feedback	PC10 and D7-D10 open responses	Negative cases retained; no automated thematic claims

Table G3. Post-condition items PC1-PC11

ID	What it recorded	Format
PC1	Completed the review and final record	Yes/No
PC2	Completed without substantial help	Yes/No
PC3	Overall task ease	1-7
PC4	Confidence in final review decisions	1-7
PC5	Ease of checking transcript sources	1-7
PC6	Clarity of evidence versus missing information	1-7
PC7	Confidence that the record reflected decisions	1-7
PC8	Mental effort	1-7
PC9	Technical interruption	Yes/No
PC10	Most difficult or useful aspect	Open text
PC11	Administrator-timed active review	Minutes

Table G4. Final comparison items D1-D10

ID	What it recorded	Format
D1	Preferred condition for the assigned task	Choice
D2	Reason for that preference	Open text
D3	Intended real-project mode	Current / Prototype / Hybrid / Neither
D4	Willingness to use Prototype	1-7
D5	Planned frequency of full-transcript checking	Choice
D6	Confidence in the reviewed Evidence Record	1-7
D7	Conditions required for adoption	Open text
D8	Reasons to refuse or stop use	Open text
D9	First improvement priority	Open text
D10	Where Prototype added most value	Open text

Appendix H. A/B Rating Rubric and Table Provenance

The A/B form used a project-specific anchored ordinal scale rather than a validated psychometric instrument. The midpoint represented partial or mixed fulfilment. Scores 2 and 4 represented intermediate cases between the stated anchors. Overall preference (A, B or Tie) remained a separate categorical judgement.

Table H1. Criterion-specific anchors for the 1-5 A/B rubric

Criterion	1 - clear failure	3 - partial or mixed	5 - full fulfilment
Q1 Protocol coverage	Multiple required elements absent, contradicted or completed by unsupported inference	Core topic partly covered, with a material omission or ambiguity	Required protocol elements are covered without unsupported completion
Q2 Probing relevance and neutrality	Probe is irrelevant, leading, repetitive or outside the permitted purpose	Probe is partly targeted but broad, awkward or mildly leading	Probe is necessary, focused, neutral and within the bounded purpose
Q3 Transcript grounding and fidelity	Important content is unsupported or changes the participant's meaning	Core content is grounded but one claim is broad, weakly linked or over-interpreted	Important claims are traceable to relevant source and preserve meaning
Q4 Participant clarity and control	Skip, Stop, refusal or boundary is ignored, obscured or pressured	Control is visible but consequences or wording are inconsistent	Participant control is clear, respected and reflected downstream
Q5 Researcher reviewability	Source, limitations or available review actions are difficult to locate or interpret	Record is usable but contains avoidable ambiguity or checking burden	Source, draft, limitations and review actions are clear and inspectable

Table H2 records how the report's principal tables were produced. It distinguishes automated exports from human coding and keeps each denominator tied to its source material.

Table H2. Data provenance for principal report tables

Report output	Raw inputs	Generation method	Human / automated role	Claim boundary
Consultation findings	Three included de-identified written forms	Framework matrix: current problem, need/tension and design implication	Author coding; short quotations retained	Formative needs evidence, not usability or saturation
Requirements and success criteria	Literature, consultation themes, prototype failures and ethics boundary	Need-to-design and requirement-to-evidence mapping	Author design synthesis	Requirements are justified design propositions, not stakeholder votes
Formal MVP / Baseline results	32 registered frozen attempts, raw calls and validator outputs	Runner exports and independent oracle validation	Automated generation with retained failures	Fixed scripted trajectory; no natural-session or SOTA claim
Masked A/B ratings	Ten complete eligible pairs and two locked coder returns	Independent anchored ratings; means over applicable coder-pair observations	Human coding; no consensus round	Descriptive paired result, not target-user preference
Supplementary MP	Nine final-reviewed MVP items / five unique bundles	Two independent Preserved / Not preserved labels	Human coding	Within-MVP fidelity check; no Baseline comparison
Supplementary UCR	Frozen 46 propositions / 27 unique texts from the same nine items	Independent support labels, initial agreement, one resolved disagreement	Human coding; administrator-frozen inventory	No Baseline UCR and no coder-consensus segmentation claim
Researcher-review task	Six final condition returns, three Evidence Records, Gold v0.8.0 and response workbooks	60 Gold-aligned decisions plus descriptive medians, choices and paraphrased open responses	Human task completion; administrator-generated descriptive analysis	Exploratory n=3 evidence, including objective stage timing; no inferential or whole-workflow claim

Appendix I. Researcher Feedback and Post-validation Design Directions

The main text reports the task outcome and three representative comments. Table I1 preserves the cross-participant design implications without treating three responses as saturation or prevalence. Requested changes were not added to the evaluated build.

Table I1. Qualitative findings, implications and implementation status

Finding	Evidence across P01-P03	Design implication	Status
Stable source anchors	All required dependable links or source-anchor checks	Retain direct anchors and add routine anchor-quality sampling	Implemented; more validation needed
Full transcript remains primary	All described returning to the participant's own words or full transcript	Keep transcript navigation and neighbouring context	Implemented
Researcher authority	P01 required the researcher to make the final call; P03 framed a pre-sign-off control	Keep Include/Edit/Exclude and approval gates	Implemented
Decision history	Participants required an intelligible edit and provenance trail	Preserve actor, time, prior state and reopened decisions	Implemented
Topical relevance is not support	P02 and P03 rejected same-topic or lexical similarity as substantiation	Keep semantic support checks separate from link validation	Implemented conceptually
Distinct failure labels	P02 and P03 requested discrepancy-specific feedback	Separate unsupported assertion from coverage gap	Future work; not evaluated
Quotation and stable ordering	P01 reported ordering effort; P02/P03 requested source text at the decision point	Show attributed text beside the proposition without changing evidence IDs	Future work; not evaluated

Appendix J. Supplementary Implementation Views

These views document a bounded end-to-end vertical slice from researcher setup and participant text collection to review and HTML Evidence Record handoff. They do not cover recruitment, scheduling, qualitative coding, report writing or production deployment. Each caption states the requirement represented and the limit of what the screenshot can prove.

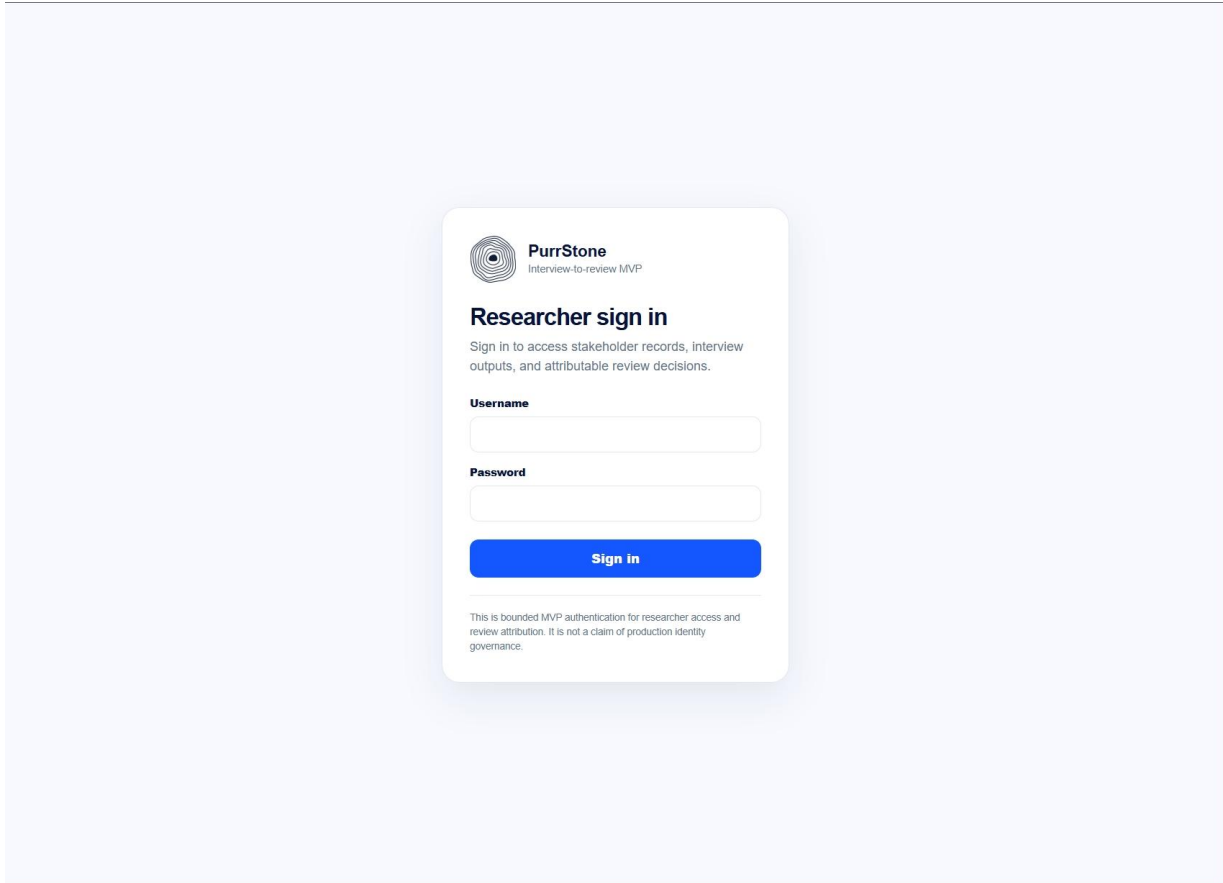


Figure J1. Researcher sign-in. Workflow evidence: R5 role separation. This view documents the implemented access screen, not production-grade authentication or security.

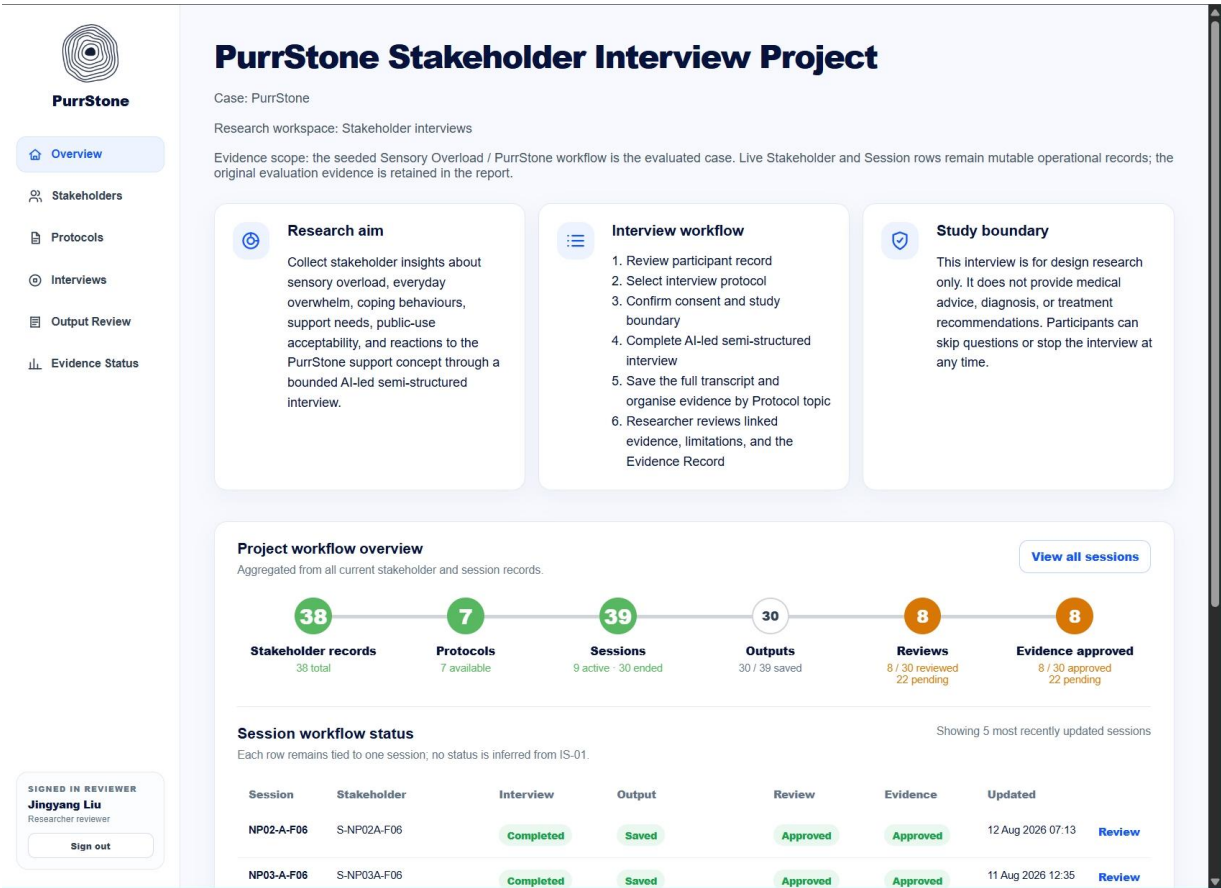


Figure J2. Researcher overview. Workflow evidence: R1 setup and workflow status. This view documents navigation and saved session status, not ease of use.

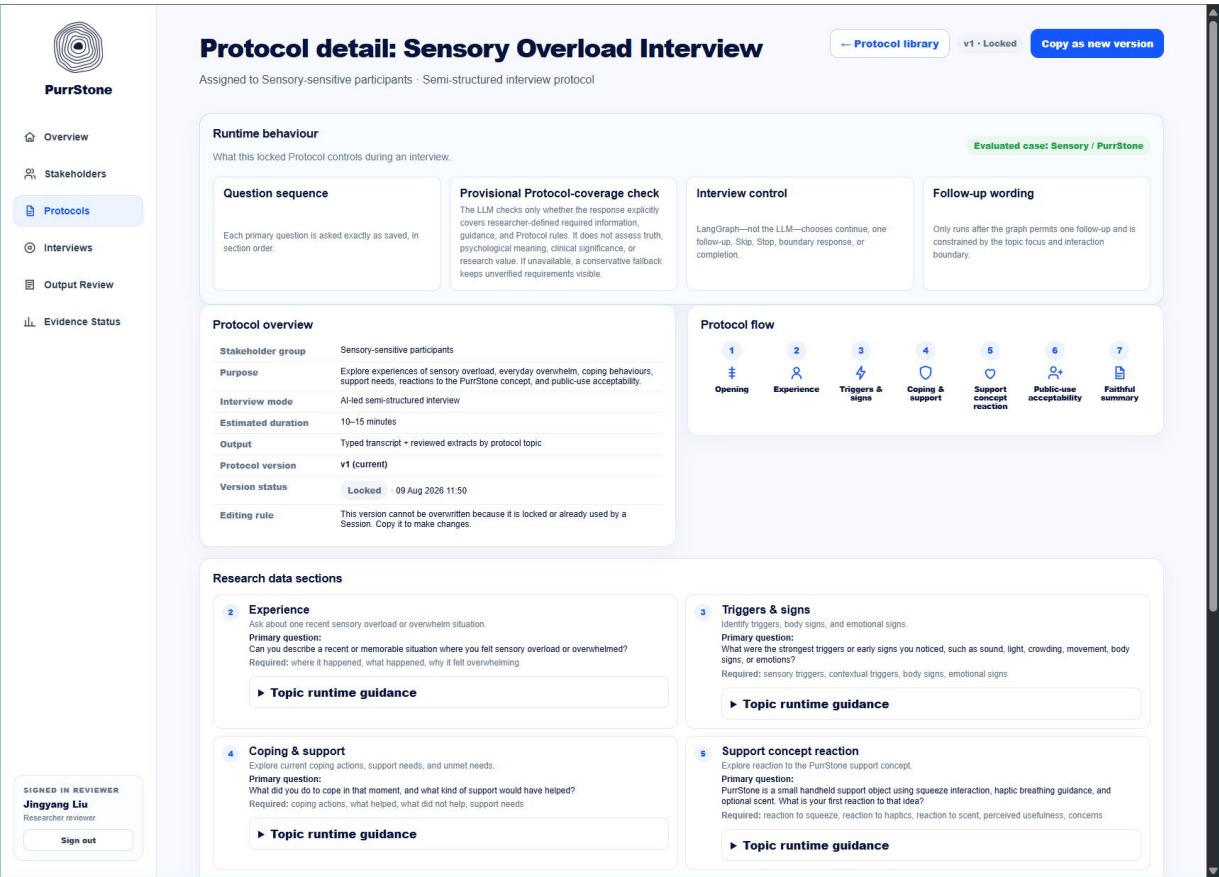


Figure J3. Protocol detail and configuration. Workflow evidence: R1 purpose, questions, required information and probe limit. This view documents implemented fields, not the quality of the interview guide.

PurrStone

PurrStone Interview

Sensory Overload Interview

AI-led semi-structured interview for Sensory-sensitive participants.

- 1. What this interview is about**

Explore experiences of sensory overload, everyday overwhelm, coping behaviours, support needs, reactions to the PurrStone concept, and public-use acceptability.
Protocol topics: Experience, Triggers & signs, Coping & support, Support concept reaction, Public-use acceptability.
- 2. How the AI-supported interview works**
 - The system makes a provisional check of whether your responses explicitly cover information that the researcher defined for each Protocol topic.
 - It may generate and ask up to one follow-up question for each interview topic.
 - It does not assess whether your account is true. Interpret its psychological meaning, determine clinical significance, or judge its research value.
 - The provisional coverage check may determine whether the interview moves to the next topic or asks that one follow-up.
 - Any material included in the final Evidence Record must be reviewed and approved by a researcher.
- 3. Study boundary**

This interview is for design research only. It does not provide medical advice, diagnosis, treatment recommendations, or a research conclusion.
- 4. Your rights as a participant**
 - You can skip any question.
 - You can stop the interview at any time.
 - Please do not share sensitive personal information that you are not comfortable having saved in the typed transcript.
 - Your typed responses will be saved for researcher review.
- 5. Before you begin**
 - Estimated time: 10–15 minutes
 - Format: AI-led semi-structured interview
 - Participant group: Sensory-sensitive participants
 - Research output: Typed transcript + reviewed extracts by protocol topic

☐ I understand the purpose, AI-supported process, limits, participant controls, and researcher-review boundary described above, and I want to continue.

Notice version: purrstone-ai-transparency-v1

Start interview

Figure J4. Participant consent. Workflow evidence: R2 study boundary, AI disclosure, Skip and Stop. This view documents the presented consent route, not comprehension in a representative sample.

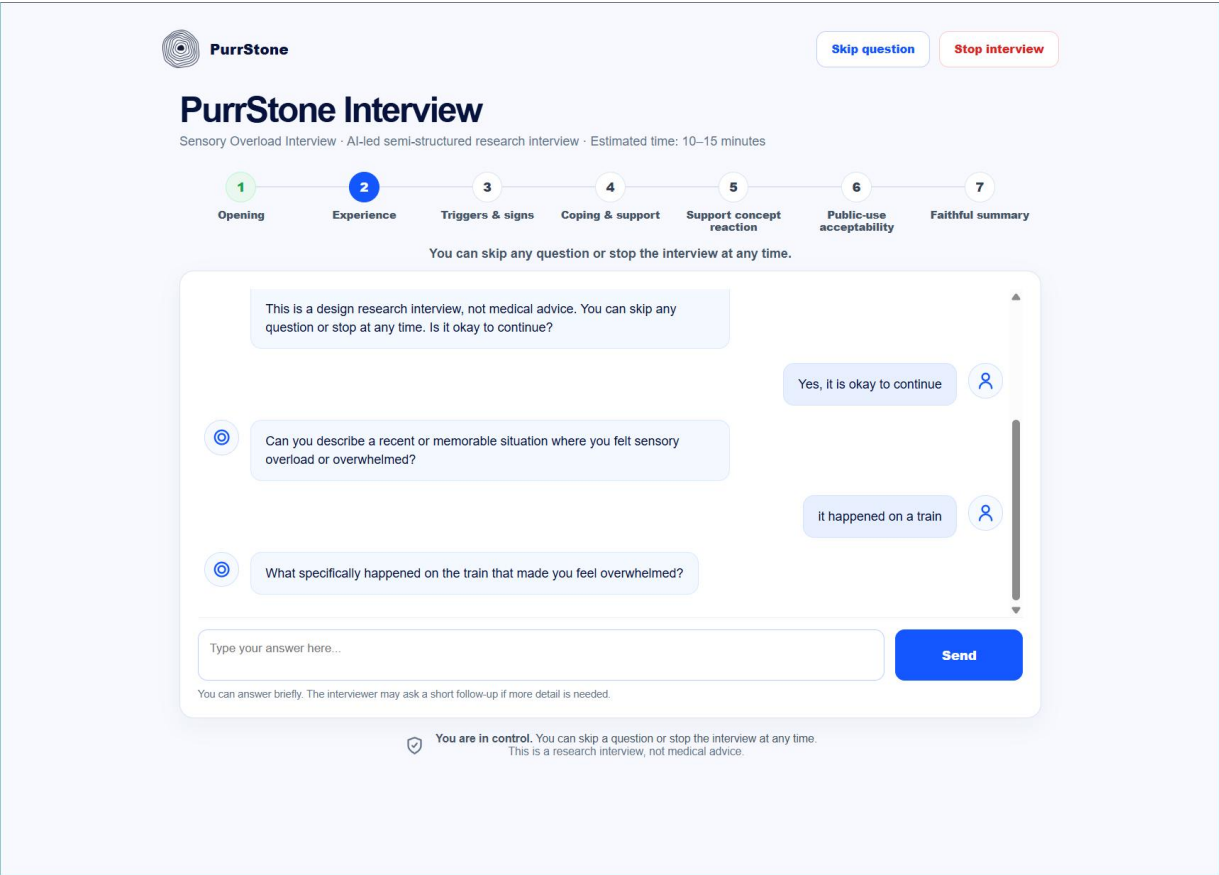


Figure J5. Participant interview. Workflow evidence: R1 bounded progression and R2 Skip/Stop controls. This view documents the text interface, not natural interview quality or rapport.

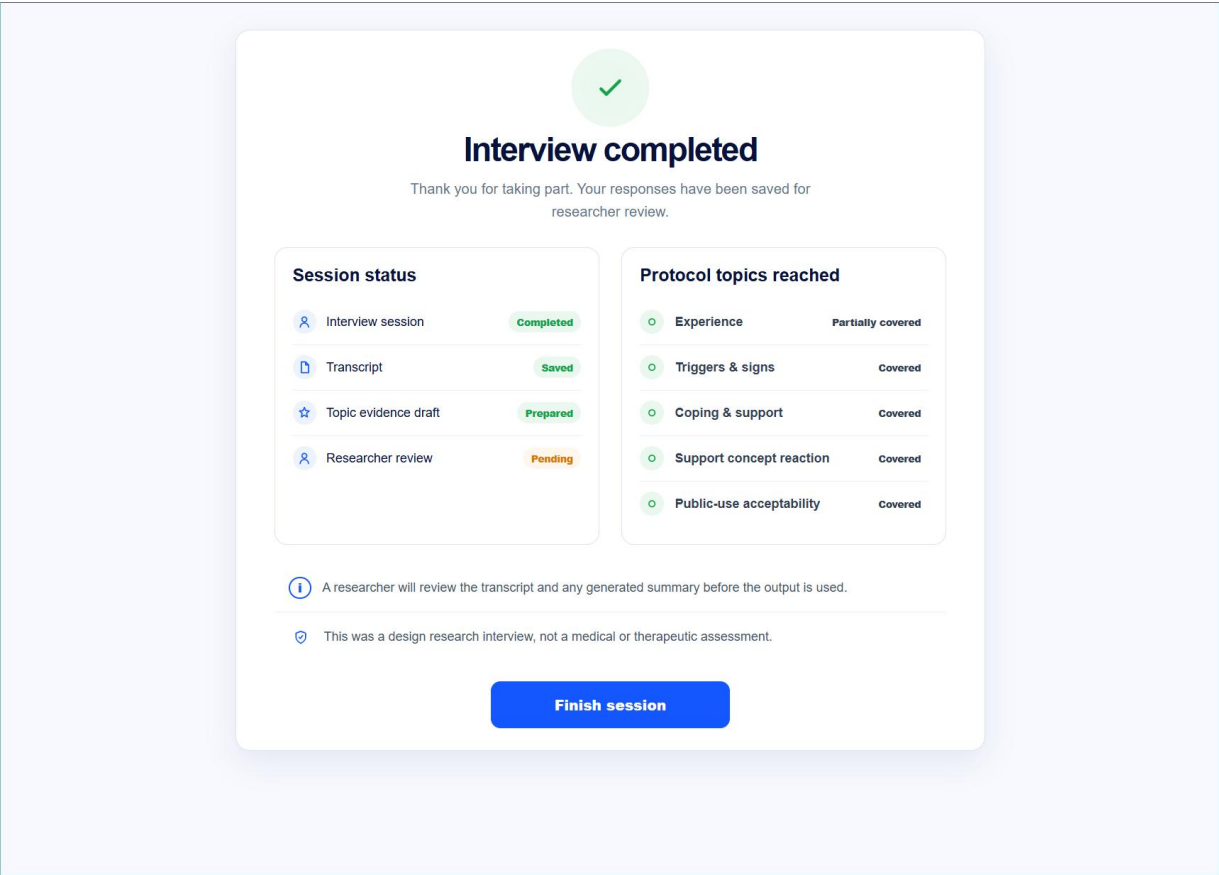


Figure J6. Interview completion. Workflow evidence: R4 topic states and visible incompleteness. This view documents stored completion states, not that every state is semantically correct.



PurrStone

Overview

Stakeholders

Protocols

Interviews

Output Review

Evidence Status

Triggers & signs

MAIN TRIGGERS

Covered

Edited

SOURCE EXTRACT FROM TYPED RESPONSES

Typed response extract: "The sharp braking noise, flashing carriage lights, and crowding were the strongest triggers. My shoulders tightened, my heart raced, and I felt panicked."

CURRENT LINKED TRANSCRIPT SOURCE(S)

Participant response 4

Verify or correct source, coverage, and missingness

Select the participant response(s) that actually support this candidate. Clear all sources when no response supports it.

Typed participant responses in this Session

☐ Participant response 2 • Experience
It happened on a train.

☐ Participant response 3 • Experience
I do not remember anything else about it.

☒ Participant response 4 • Triggers & signs
The sharp braking noise, flashing carriage lights, and crowding were the strongest triggers. My shoulders tightened, my heart raced, and I felt panicked.

☐ Participant response 5 • Coping & support
I covered my ears, focused on slow breathing, and left at the next stop. A quieter carriage or a warning before the alarm would have helped. The repeated announcements did not help because they added more noise.

☐ Participant response 6 • Support concept reaction
The sequence interaction seems useful because it would give my hands a focus, and gentle haptics might help me pace my breathing. I would avoid scent because it can be another trigger, and I would want every feature to be optional.

☐ Participant response 7 • Public-use acceptability
I might use it while commuting if it looked discreet and made no sound. I would be embarrassed by obvious lights, and scent would be a concern for people nearby in a shared space.

Coverage status for this candidate

Covered

Missing information (one item per line; blank when fully covered)

Reason for correction

Required only when you change a source, coverage status, or missing-information record.

Save source / coverage correction

SIGNED IN REVIEWER

Jingyang Liu

Researcher reviewer

Sign out

Figure J7. Item-level researcher review. Workflow evidence: R3 linked source and R5 Include/Edit/Exclude. This view documents the checking controls; a valid link does not guarantee support.

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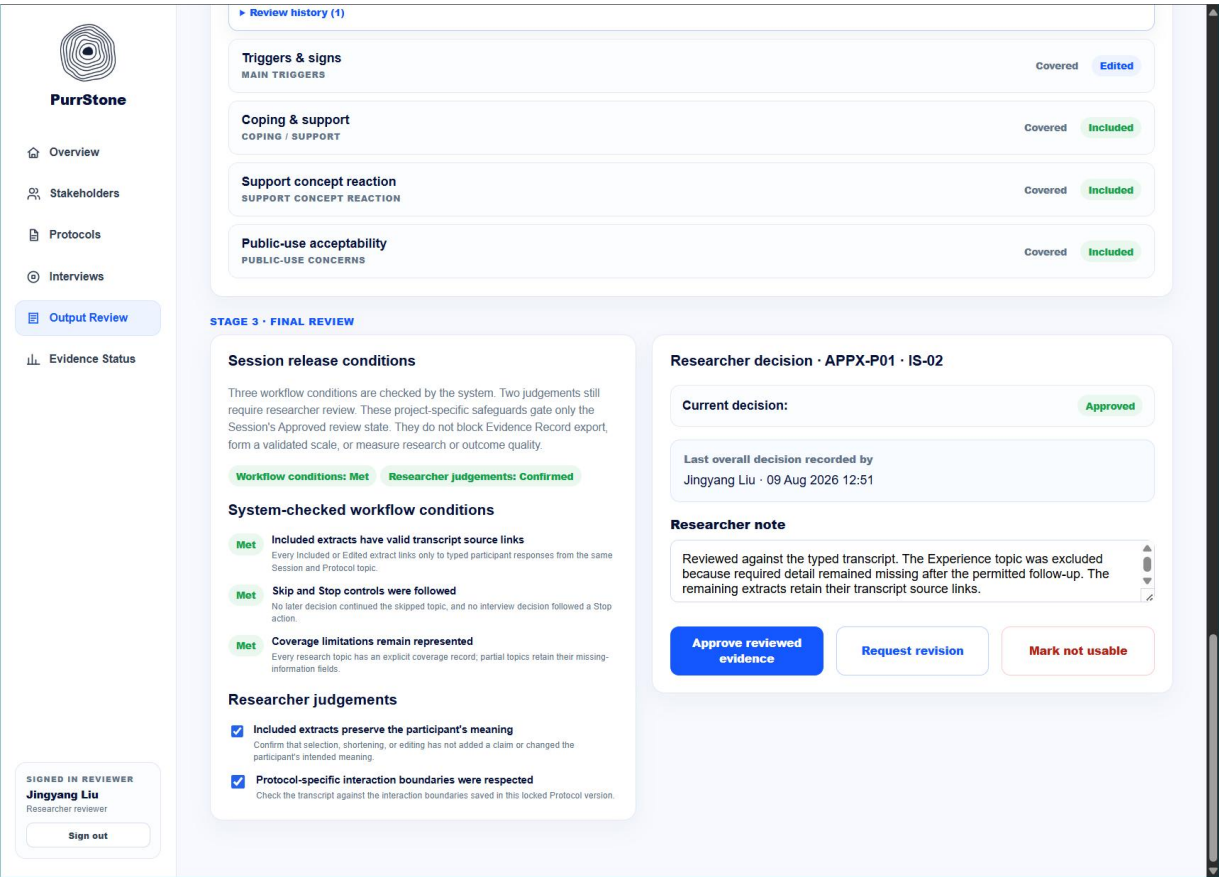


Figure J8. Overall researcher decision. Workflow evidence: R5 approval, revision request and reviewer note. This view documents the sign-off mechanism, not that the human decision is correct.

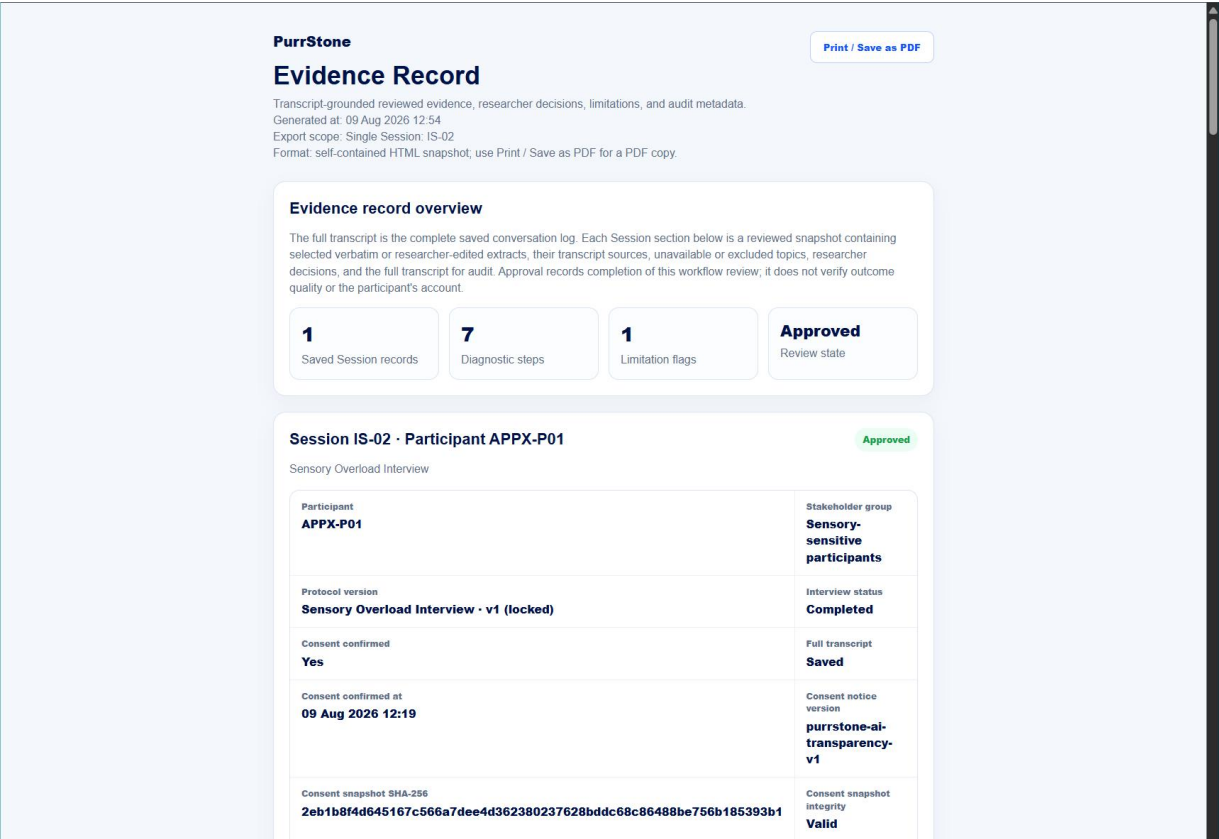


Figure J9. Final Evidence Record HTML view. Workflow evidence: R3-R5 reviewed source, state and handoff. This view documents the HTML render and print/save control; a validated PDF export was not archived.